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United States Patent

12405471

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

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Apparatus for providing waveguide displays with two-dimensional pupil expansion

Abstract

An optical display comprises: a first waveguide comprising a first surface and a second surface, an input coupler, a fold grating, and an output grating. The input coupler receives collimated first wavelength light from an Input Image Node causes the light to travel within the first waveguide via total internal reflection between the first surface and the second surface to the fold grating. The fold grating provides pupil expansion in a first direction directs the light to the output grating via total internal reflection between the first surface and the second surface. The output grating provides pupil expansion in a second direction different than the first direction and causes the light to exit the first waveguide from the first surface or the second surface. At least one of the input coupler, fold grating and output grating is a rolled k-vector grating, and the fold grating is a dual interaction grating.

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Appl. No.: 18/465091

Filed: September 11, 2023

Prior Publication Data

Document Identifier

US 20240255760 A1

Publication Date

Aug. 01, 2024

Related U.S. Application Data

continuation parent-doc US 17655756 20220321 US 11754842 child-doc US 18465091
continuation parent-doc US 17118285 20201210 US 11281013 20220322 child-doc US 17655756
continuation parent-doc US 16806947 20200302 ABANDONED child-doc US 17118285
continuation parent-doc US 15765243 US 10690916 20200623 WO PCT/GB2016/000181
20161004 child-doc US 16806947
us-provisional-application US 62285275 20151023
us-provisional-application US 62284603 20151005

Publication Classification

Int. Cl.: **G02B27/01** (20060101); **F21V8/00** (20060101); **G02B5/18** (20060101); **G02B27/00** (20060101); **G02F1/29** (20060101)

U.S. Cl.:

CPC **G02B27/0172** (20130101); **G02B5/1866** (20130101); **G02B6/0016** (20130101); **G02B6/0035** (20130101); **G02B6/0076** (20130101); **G02B27/0081** (20130101); **G02B27/0093** (20130101); **G02B27/0179** (20130101); **G02F1/292** (20130101); **G02B2027/0114** (20130101); **G02B2027/0125** (20130101); **G02B2027/0187** (20130101)

Field of Classification Search

CPC: G02B (27/0172); G02B (5/1866); G02B (6/0016); G02B (6/0035); G02B (6/0076); G02B (27/0081); G02B (27/0093); G02B (27/0179); G02B (2027/0114); G02B (2027/0125); G02B (2027/0187); G02B (27/017); G02B (6/005); G02B (2027/0178); G06F (3/0421); G06F (3/0433); G06F (3/0436); G06F (3/044); G02F (1/292)

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6069728	12/1999	Huignard et al.	N/A	N/A
6075626	12/1999	Mizutani et al.	N/A	N/A
6078427	12/1999	Fontaine et al.	N/A	N/A

6084998	12/1999	Straayer	N/A	N/A
6094311	12/1999	Moskovich	N/A	N/A
6097551	12/1999	Kreitzer	N/A	N/A
6104448	12/1999	Doane et al.	N/A	N/A
6107943	12/1999	Schroeder	N/A	N/A
6115152	12/1999	Popovich et al.	N/A	N/A
6118908	12/1999	Bischel et al.	N/A	N/A
6121899	12/1999	Theriault	N/A	N/A
6124954	12/1999	Popovich et al.	N/A	N/A
6127066	12/1999	Ueda et al.	N/A	N/A
6128058	12/1999	Walton et al.	N/A	N/A
6133971	12/1999	Silverstein et al.	N/A	N/A
6133975	12/1999	Li et al.	N/A	N/A
6137630	12/1999	Tsou et al.	N/A	N/A
6141074	12/1999	Bos et al.	N/A	N/A
6141154	12/1999	Kreitzer et al.	N/A	N/A
6151142	12/1999	Phillips et al.	N/A	N/A
6154190	12/1999	Yang et al.	N/A	N/A
6156243	12/1999	Kosuga et al.	N/A	N/A
6167169	12/1999	Brinkman et al.	N/A	N/A
6169594	12/2000	Aye et al.	N/A	N/A
6169613	12/2000	Amitai et al.	N/A	N/A
6169636	12/2000	Kreitzer et al.	N/A	N/A
6172792	12/2000	Jepsen et al.	N/A	N/A
6176837	12/2000	Foxlin	N/A	N/A
6185015	12/2000	Reinhorn et al.	N/A	N/A
6185016	12/2000	Popovich	N/A	N/A
6188462	12/2000	Lavrentovich et al.	N/A	N/A
6191887	12/2000	Michaloski et al.	N/A	N/A
6195206	12/2000	Yona et al.	N/A	N/A
6195209	12/2000	Kreitzer et al.	N/A	N/A
6204835	12/2000	Yang et al.	N/A	N/A
6211976	12/2000	Popovich et al.	N/A	N/A
6215579	12/2000	Bloom et al.	N/A	N/A
6218316	12/2000	Marsh	N/A	N/A
6222297	12/2000	Perdue	N/A	N/A
6222675	12/2000	Mall et al.	N/A	N/A
6222971	12/2000	Veligdan et al.	N/A	N/A
6249386	12/2000	Yona et al.	N/A	N/A
6259423	12/2000	Tokito et al.	N/A	N/A
6259559	12/2000	Kobayashi et al.	N/A	N/A
6266166	12/2000	Katsumata et al.	N/A	N/A
6268839	12/2000	Yang et al.	N/A	N/A
6269203	12/2000	Davies et al.	N/A	N/A
6275031	12/2000	Simmonds et al.	N/A	N/A
6278429	12/2000	Ruth et al.	N/A	N/A
6281457	12/2000	Chang	N/A	N/A
6285813	12/2000	Schultz et al.	N/A	N/A
6297860	12/2000	Moskovich et al.	N/A	N/A
6301056	12/2000	Kreitzer et al.	N/A	N/A

6301057	12/2000	Kreitzer et al.	N/A	N/A
6317083	12/2000	Johnson et al.	N/A	N/A
6317189	12/2000	Yuan et al.	N/A	N/A
6317227	12/2000	Mizutani et al.	N/A	N/A
6317228	12/2000	Popovich et al.	N/A	N/A
6317528	12/2000	Gadkaree et al.	N/A	N/A
6320563	12/2000	Yang et al.	N/A	N/A
6321069	12/2000	Piirainen	N/A	N/A
6323970	12/2000	Popovich	N/A	N/A
6323989	12/2000	Jacobson et al.	N/A	N/A
6324014	12/2000	Moskovich et al.	N/A	N/A
6327089	12/2000	Hosaki et al.	N/A	N/A
6330109	12/2000	Ishii et al.	N/A	N/A
6333819	12/2000	Svedenkrans	N/A	N/A
6335224	12/2001	Peterson et al.	N/A	N/A
6339486	12/2001	Popovich	N/A	N/A
6340540	12/2001	Ueda et al.	N/A	N/A
6351273	12/2001	Lemelson et al.	N/A	N/A
6351333	12/2001	Araki et al.	N/A	N/A
6356172	12/2001	Koivisto et al.	N/A	N/A
6356674	12/2001	Davis et al.	N/A	N/A
6359730	12/2001	Tervonen	N/A	N/A
6359737	12/2001	Stringfellow	N/A	N/A
6366281	12/2001	Lipton et al.	N/A	N/A
6366369	12/2001	Ichikawa et al.	N/A	N/A
6366378	12/2001	Tervonen et al.	N/A	N/A
6377238	12/2001	McPheters	N/A	N/A
6377321	12/2001	Khan et al.	N/A	N/A
6388797	12/2001	Lipton et al.	N/A	N/A
6392812	12/2001	Howard	N/A	N/A
6407724	12/2001	Waldern et al.	N/A	N/A
6409687	12/2001	Foxlin	N/A	N/A
6411444	12/2001	Moskovich et al.	N/A	N/A
6414760	12/2001	Lopez et al.	N/A	N/A
6417971	12/2001	Moskovich et al.	N/A	N/A
6421109	12/2001	Popovich	N/A	N/A
6437563	12/2001	Simmonds et al.	N/A	N/A
6437886	12/2001	Trepanier et al.	N/A	N/A
6445512	12/2001	Moskovich et al.	N/A	N/A
6449095	12/2001	Ohtaki et al.	N/A	N/A
6456584	12/2001	Nagata et al.	N/A	N/A
6470132	12/2001	Nousiainen et al.	N/A	N/A
6473209	12/2001	Popovich	N/A	N/A
6476974	12/2001	Kreitzer et al.	N/A	N/A
6483303	12/2001	Simmonds et al.	N/A	N/A
6486997	12/2001	Bruzzzone et al.	N/A	N/A
6504518	12/2002	Kuwayama et al.	N/A	N/A
6504629	12/2002	Popovich et al.	N/A	N/A
6509937	12/2002	Moskovich et al.	N/A	N/A
6510263	12/2002	Maisenhoelder et al.	N/A	N/A

6518747	12/2002	Sager et al.	N/A	N/A
6519088	12/2002	Lipton	N/A	N/A
6522794	12/2002	Bischel et al.	N/A	N/A
6522795	12/2002	Jordan et al.	N/A	N/A
6524771	12/2002	Maeda et al.	N/A	N/A
6529336	12/2002	Kreitzer et al.	N/A	N/A
6534977	12/2002	Duncan et al.	N/A	N/A
6538775	12/2002	Bowley et al.	N/A	N/A
6545778	12/2002	Ono et al.	N/A	N/A
6545808	12/2002	Ehbets et al.	N/A	N/A
6550949	12/2002	Bauer et al.	N/A	N/A
6552789	12/2002	Modro	N/A	N/A
6557413	12/2002	Nieminen et al.	N/A	N/A
6559813	12/2002	DeLuca et al.	N/A	N/A
6560019	12/2002	Nakai	N/A	N/A
6563648	12/2002	Gleckman et al.	N/A	N/A
6563650	12/2002	Moskovich et al.	N/A	N/A
6567014	12/2002	Hansen et al.	N/A	N/A
6567573	12/2002	Domash et al.	N/A	N/A
6577411	12/2002	David et al.	N/A	N/A
6577429	12/2002	Kurtz et al.	N/A	N/A
6580529	12/2002	Amitai et al.	N/A	N/A
6583838	12/2002	Hoke et al.	N/A	N/A
6583873	12/2002	Goncharov et al.	N/A	N/A
6587269	12/2002	Li	N/A	N/A
6587619	12/2002	Kinoshita	N/A	N/A
6594090	12/2002	Kruschwitz et al.	N/A	N/A
6596193	12/2002	Coates et al.	N/A	N/A
6597176	12/2002	Simmonds et al.	N/A	N/A
6597475	12/2002	Shirakura et al.	N/A	N/A
6598987	12/2002	Parikka	N/A	N/A
6600590	12/2002	Roddy et al.	N/A	N/A
6608720	12/2002	Freeman	N/A	N/A
6611253	12/2002	Cohen	N/A	N/A
6618104	12/2002	Date et al.	N/A	N/A
6624943	12/2002	Nakai et al.	N/A	N/A
6625381	12/2002	Roddy et al.	N/A	N/A
6646772	12/2002	Popovich et al.	N/A	N/A
6646810	12/2002	Harter, Jr. et al.	N/A	N/A
6661495	12/2002	Popovich	N/A	N/A
6661578	12/2002	Hedrick	N/A	N/A
6667134	12/2002	Sutherland et al.	N/A	N/A
6674578	12/2003	Sugiyama et al.	N/A	N/A
6677086	12/2003	Sutehrland et al.	N/A	N/A
6678093	12/2003	Scobey et al.	N/A	N/A
6680720	12/2003	Lee et al.	N/A	N/A
6686815	12/2003	Mirshekarl-Syahkal et al.	N/A	N/A
6690516	12/2003	Aritake et al.	N/A	N/A
6692666	12/2003	Sutherland et al.	N/A	N/A

6699407	12/2003	Sutehrland et al.	N/A	N/A
6706086	12/2003	Emig et al.	N/A	N/A
6706451	12/2003	Sutherland et al.	N/A	N/A
6714329	12/2003	Sekine et al.	N/A	N/A
6721096	12/2003	Bruzzozone et al.	N/A	N/A
6730442	12/2003	Sutherland et al.	N/A	N/A
6731434	12/2003	Hua et al.	N/A	N/A
6738105	12/2003	Hannah et al.	N/A	N/A
6741189	12/2003	Gibbons, II et al.	N/A	N/A
6744478	12/2003	Asakura et al.	N/A	N/A
6747781	12/2003	Trisnadi et al.	N/A	N/A
6748342	12/2003	Dickhaus	N/A	N/A
6750941	12/2003	Satoh et al.	N/A	N/A
6750995	12/2003	Dickson	N/A	N/A
6750996	12/2003	Jagt et al.	N/A	N/A
6757105	12/2003	Niv et al.	N/A	N/A
6771403	12/2003	Endo et al.	N/A	N/A
6776339	12/2003	Piikivi	N/A	N/A
6781701	12/2003	Sweetser et al.	N/A	N/A
6791629	12/2003	Moskovich et al.	N/A	N/A
6791739	12/2003	Ramanujan et al.	N/A	N/A
6804066	12/2003	Ha et al.	N/A	N/A
6805490	12/2003	Levola	N/A	N/A
6821457	12/2003	Natarajan et al.	N/A	N/A
6822713	12/2003	Yaroshchuk et al.	N/A	N/A
6824929	12/2003	Taggi et al.	N/A	N/A
6825987	12/2003	Repetto et al.	N/A	N/A
6829095	12/2003	Amitai	N/A	N/A
6830789	12/2003	Doane et al.	N/A	N/A
6833955	12/2003	Niv	N/A	N/A
6836369	12/2003	Fujikawa et al.	N/A	N/A
6842563	12/2004	Zhang et al.	N/A	N/A
6844212	12/2004	Bond et al.	N/A	N/A
6844980	12/2004	He et al.	N/A	N/A
6844989	12/2004	Jo et al.	N/A	N/A
6847274	12/2004	Salmela et al.	N/A	N/A
6847488	12/2004	Travis	N/A	N/A
6850210	12/2004	Lipton et al.	N/A	N/A
6853491	12/2004	Ruhle et al.	N/A	N/A
6853493	12/2004	Kreitzer et al.	N/A	N/A
6861107	12/2004	Klasen-memmer et al.	N/A	N/A
6864861	12/2004	Schehrer et al.	N/A	N/A
6864927	12/2004	Cathey	N/A	N/A
6864931	12/2004	Kumar et al.	N/A	N/A
6867888	12/2004	Sutherland et al.	N/A	N/A
6873443	12/2004	Joubert et al.	N/A	N/A
6876791	12/2004	Murashima et al.	N/A	N/A
6878494	12/2004	Sutehrland et al.	N/A	N/A
6885483	12/2004	Takada	N/A	N/A

6903872	12/2004	Schrader	N/A	N/A
6909345	12/2004	Salmela et al.	N/A	N/A
6917375	12/2004	Akada et al.	N/A	N/A
6919003	12/2004	Ikeda et al.	N/A	N/A
6922267	12/2004	Endo et al.	N/A	N/A
6926429	12/2004	Barlow et al.	N/A	N/A
6927570	12/2004	Simmonds et al.	N/A	N/A
6927694	12/2004	Smith et al.	N/A	N/A
6940361	12/2004	Jokio et al.	N/A	N/A
6943788	12/2004	Tomono	N/A	N/A
6950173	12/2004	Sutherland et al.	N/A	N/A
6950227	12/2004	Schrader	N/A	N/A
6951393	12/2004	Koide	N/A	N/A
6952312	12/2004	Weber et al.	N/A	N/A
6952435	12/2004	Lai et al.	N/A	N/A
6958662	12/2004	Salmela et al.	N/A	N/A
6958868	12/2004	Pender	N/A	N/A
6963454	12/2004	Martins et al.	N/A	N/A
6972788	12/2004	Robertson et al.	N/A	N/A
6975345	12/2004	Lipton et al.	N/A	N/A
6980365	12/2004	Moskovich	N/A	N/A
6985296	12/2005	Lipton et al.	N/A	N/A
6987908	12/2005	Bond et al.	N/A	N/A
6999239	12/2005	Martins et al.	N/A	N/A
7002618	12/2005	Lipton et al.	N/A	N/A
7002753	12/2005	Moskovich et al.	N/A	N/A
7003075	12/2005	Miyake et al.	N/A	N/A
7003187	12/2005	Frick et al.	N/A	N/A
7006732	12/2005	Gunn, III et al.	N/A	N/A
7009773	12/2005	Chaoulov et al.	N/A	N/A
7018563	12/2005	Sutherland et al.	N/A	N/A
7018686	12/2005	Sutehrland et al.	N/A	N/A
7018744	12/2005	Otaki et al.	N/A	N/A
7019793	12/2005	Moskovich et al.	N/A	N/A
7021777	12/2005	Amitai	N/A	N/A
7026892	12/2005	Kajiya	N/A	N/A
7027671	12/2005	Huck et al.	N/A	N/A
7034748	12/2005	Kajiya	N/A	N/A
7046439	12/2005	Kaminsky et al.	N/A	N/A
7050674	12/2005	Lee et al.	N/A	N/A
7053735	12/2005	Salmela et al.	N/A	N/A
7053991	12/2005	Sandusky	N/A	N/A
7054045	12/2005	McPheters et al.	N/A	N/A
7058434	12/2005	Wang et al.	N/A	N/A
7068405	12/2005	Sutherland et al.	N/A	N/A
7068898	12/2005	Buretea et al.	N/A	N/A
7072020	12/2005	Sutherland et al.	N/A	N/A
7075273	12/2005	O'Gorman et al.	N/A	N/A
7077984	12/2005	Natarajan et al.	N/A	N/A
7081215	12/2005	Natarajan et al.	N/A	N/A

7088457	12/2005	Zou et al.	N/A	N/A
7088515	12/2005	Lipton	N/A	N/A
7095562	12/2005	Peng et al.	N/A	N/A
7099080	12/2005	Lipton et al.	N/A	N/A
7101048	12/2005	Travis	N/A	N/A
7108383	12/2005	Mitchell et al.	N/A	N/A
7110184	12/2005	Yona et al.	N/A	N/A
7119965	12/2005	Rolland et al.	N/A	N/A
7123418	12/2005	Weber et al.	N/A	N/A
7123421	12/2005	Moskovich et al.	N/A	N/A
7126418	12/2005	Hunton et al.	N/A	N/A
7126583	12/2005	Breed	N/A	N/A
7132200	12/2005	Ueda et al.	N/A	N/A
7133084	12/2005	Moskovich et al.	N/A	N/A
7139109	12/2005	Mukawa	N/A	N/A
RE39424	12/2005	Moskovich	N/A	N/A
7145729	12/2005	Kreitzer et al.	N/A	N/A
7149385	12/2005	Parikka et al.	N/A	N/A
7151246	12/2005	Fein et al.	N/A	N/A
7158095	12/2006	Jenson et al.	N/A	N/A
7167286	12/2006	Anderson et al.	N/A	N/A
7167616	12/2006	Ling et al.	N/A	N/A
7175780	12/2006	Sutherland et al.	N/A	N/A
7181105	12/2006	Teramura et al.	N/A	N/A
7181108	12/2006	Levola	N/A	N/A
7184002	12/2006	Lipton et al.	N/A	N/A
7184615	12/2006	Levola	N/A	N/A
7186567	12/2006	Sutherland et al.	N/A	N/A
7190849	12/2006	Katase	N/A	N/A
7198737	12/2006	Natarajan et al.	N/A	N/A
7199934	12/2006	Yamasaki	N/A	N/A
7205960	12/2006	David	N/A	N/A
7205964	12/2006	Yokoyama et al.	N/A	N/A
7206107	12/2006	Levola	N/A	N/A
7212175	12/2006	Magee et al.	N/A	N/A
7218817	12/2006	Magnusson et al.	N/A	N/A
7230767	12/2006	Walck et al.	N/A	N/A
7230770	12/2006	Kreitzer et al.	N/A	N/A
7242527	12/2006	Spitzer et al.	N/A	N/A
7248128	12/2006	Mattila et al.	N/A	N/A
7248765	12/2006	Lee et al.	N/A	N/A
7256915	12/2006	Sutherland et al.	N/A	N/A
7259906	12/2006	Islam	N/A	N/A
7265882	12/2006	Sutherland et al.	N/A	N/A
7265903	12/2006	Sutherland et al.	N/A	N/A
7268946	12/2006	Wang	N/A	N/A
7280722	12/2006	Temkin et al.	N/A	N/A
7285903	12/2006	Cull et al.	N/A	N/A
7286272	12/2006	Mukawa	N/A	N/A
7289069	12/2006	Ranta	N/A	N/A

RE39911	12/2006	Moskovich	N/A	N/A
7299983	12/2006	Piikivi	N/A	N/A
7301601	12/2006	Lin et al.	N/A	N/A
7312906	12/2006	Sutherland et al.	N/A	N/A
7313291	12/2006	Okhotnikov et al.	N/A	N/A
D559250	12/2007	Pombo	N/A	N/A
7319573	12/2007	Nishiyama	N/A	N/A
7320534	12/2007	Sugikawa et al.	N/A	N/A
7323275	12/2007	Otaki et al.	N/A	N/A
7333685	12/2007	Stone et al.	N/A	N/A
7336271	12/2007	Ozeki et al.	N/A	N/A
7339737	12/2007	Urey et al.	N/A	N/A
7339742	12/2007	Amitai et al.	N/A	N/A
7349612	12/2007	Nishii et al.	N/A	N/A
7356218	12/2007	Kato et al.	N/A	N/A
7356224	12/2007	Levner et al.	N/A	N/A
7369911	12/2007	Volant et al.	N/A	N/A
7375870	12/2007	Schorpp	N/A	N/A
7375886	12/2007	Lipton et al.	N/A	N/A
7376068	12/2007	Khoury	N/A	N/A
7376307	12/2007	Singh et al.	N/A	N/A
7389023	12/2007	Yeo et al.	N/A	N/A
7391573	12/2007	Amitai	N/A	N/A
7394865	12/2007	Borran et al.	N/A	N/A
7394961	12/2007	Kornilovich et al.	N/A	N/A
7395181	12/2007	Foxlin	N/A	N/A
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7401920	12/2007	Kranz et al.	N/A	N/A
7404644	12/2007	Evans et al.	N/A	N/A
7410286	12/2007	Travis	N/A	N/A
7411637	12/2007	Weiss	N/A	N/A
7413678	12/2007	Natarajan et al.	N/A	N/A
7413679	12/2007	Sutherland et al.	N/A	N/A
7415173	12/2007	Kassamakov et al.	N/A	N/A
7416818	12/2007	Sutherland et al.	N/A	N/A
7418170	12/2007	Mukawa et al.	N/A	N/A
7420733	12/2007	Natarajan et al.	N/A	N/A
7433116	12/2007	Islam	N/A	N/A
7436568	12/2007	Kuykendall, Jr.	N/A	N/A
D581447	12/2007	Yee	N/A	N/A
7447967	12/2007	Onggosanusi et al.	N/A	N/A
7453612	12/2007	Mukawa	N/A	N/A
7454103	12/2007	Parriaux	N/A	N/A
7457040	12/2007	Amitai	N/A	N/A
7466994	12/2007	Pihlaja et al.	N/A	N/A
7477206	12/2008	Cowan et al.	N/A	N/A
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7483604	12/2008	Levola	N/A	N/A

7492512	12/2008	Niv et al.	N/A	N/A
7496293	12/2008	Shamir et al.	N/A	N/A
7499217	12/2008	Cakmakci et al.	N/A	N/A
7500104	12/2008	Goland	N/A	N/A
7511891	12/2008	Messerschmidt	N/A	N/A
7513668	12/2008	Peng et al.	N/A	N/A
7522344	12/2008	Curatu et al.	N/A	N/A
7525448	12/2008	Wilson et al.	N/A	N/A
7528385	12/2008	Volodin et al.	N/A	N/A
7542210	12/2008	Chirieleison	N/A	N/A
7545429	12/2008	Travis	N/A	N/A
7550234	12/2008	Otaki et al.	N/A	N/A
7558446	12/2008	Wimberger-friedl et al.	N/A	N/A
7567372	12/2008	Schorpp	N/A	N/A
7570322	12/2008	Sutherland et al.	N/A	N/A
7570405	12/2008	Sutherland et al.	N/A	N/A
7570429	12/2008	Maliah et al.	N/A	N/A
7572555	12/2008	Takizawa et al.	N/A	N/A
7573640	12/2008	Nivon et al.	N/A	N/A
7576916	12/2008	Amitai	N/A	N/A
7577326	12/2008	Amitai	N/A	N/A
7579119	12/2008	Ueda et al.	N/A	N/A
7583423	12/2008	Sutherland et al.	N/A	N/A
7587110	12/2008	Singh et al.	N/A	N/A
7588863	12/2008	Takizawa et al.	N/A	N/A
7589900	12/2008	Powell	N/A	N/A
7589901	12/2008	DeJong et al.	N/A	N/A
7592988	12/2008	Katase	N/A	N/A
7593575	12/2008	Houle et al.	N/A	N/A
7597447	12/2008	Larson et al.	N/A	N/A
7599012	12/2008	Nakamura et al.	N/A	N/A
7600893	12/2008	Laino et al.	N/A	N/A
7602552	12/2008	Blumenfeld	N/A	N/A
7605719	12/2008	Wenger et al.	N/A	N/A
7605774	12/2008	Brandt et al.	N/A	N/A
7605882	12/2008	Sutherland et al.	N/A	N/A
7616270	12/2008	Hirabayashi et al.	N/A	N/A
7617022	12/2008	Wood et al.	N/A	N/A
7618750	12/2008	Ueda et al.	N/A	N/A
7619739	12/2008	Sutherland et al.	N/A	N/A
7619825	12/2008	Peng et al.	N/A	N/A
7629086	12/2008	Otaki et al.	N/A	N/A
7639208	12/2008	Ha et al.	N/A	N/A
7639911	12/2008	Lee et al.	N/A	N/A
7643214	12/2009	Amitai	N/A	N/A
7643225	12/2009	Tsai	N/A	N/A
7656585	12/2009	Powell et al.	N/A	N/A
7660047	12/2009	Travis et al.	N/A	N/A
7672024	12/2009	Kuan	N/A	N/A

7672055	12/2009	Amitai	N/A	N/A
7672549	12/2009	Ghosh et al.	N/A	N/A
7675021	12/2009	Lapstun	N/A	N/A
7675684	12/2009	Weissman et al.	N/A	N/A
7691248	12/2009	Ikeda et al.	N/A	N/A
7710622	12/2009	Takabayashi et al.	N/A	N/A
7710654	12/2009	Ashkenazi et al.	N/A	N/A
7711228	12/2009	Noda et al.	N/A	N/A
7724441	12/2009	Amitai	N/A	N/A
7724442	12/2009	Amitai	N/A	N/A
7724443	12/2009	Amitai	N/A	N/A
7733571	12/2009	Li	N/A	N/A
7733572	12/2009	Brown et al.	N/A	N/A
7740387	12/2009	Schultz et al.	N/A	N/A
7747113	12/2009	Mukawa et al.	N/A	N/A
7751122	12/2009	Amitai	N/A	N/A
7751662	12/2009	Kleemann et al.	N/A	N/A
7764413	12/2009	Levola	N/A	N/A
7777819	12/2009	Simmonds	N/A	N/A
7778305	12/2009	Parriaux et al.	N/A	N/A
7778508	12/2009	Hirayama	N/A	N/A
7843642	12/2009	Shaoulov et al.	N/A	N/A
7847235	12/2009	Krupkin et al.	N/A	N/A
7864427	12/2010	Korenaga et al.	N/A	N/A
7865080	12/2010	Hecker et al.	N/A	N/A
7866869	12/2010	Karakawa	N/A	N/A
7872707	12/2010	Sutherland et al.	N/A	N/A
7872804	12/2010	Moon et al.	N/A	N/A
7884593	12/2010	Simmonds et al.	N/A	N/A
7884985	12/2010	Amitai et al.	N/A	N/A
7887186	12/2010	Watanabe	N/A	N/A
7903921	12/2010	Ostergard	N/A	N/A
7907342	12/2010	Simmonds et al.	N/A	N/A
7920787	12/2010	Gentner et al.	N/A	N/A
7928862	12/2010	Matthews	N/A	N/A
7936513	12/2010	Wu et al.	N/A	N/A
7936519	12/2010	Mukawa et al.	N/A	N/A
7944428	12/2010	Travis	N/A	N/A
7944616	12/2010	Mukawa	N/A	N/A
7949214	12/2010	DeJong et al.	N/A	N/A
D640310	12/2010	Suzuki et al.	N/A	N/A
7961117	12/2010	Zimmerman et al.	N/A	N/A
7969644	12/2010	Tilleman et al.	N/A	N/A
7969657	12/2010	Cakmakci et al.	N/A	N/A
7970246	12/2010	Travis et al.	N/A	N/A
7976208	12/2010	Travis	N/A	N/A
7984884	12/2010	Iliev et al.	N/A	N/A
7999982	12/2010	Endo et al.	N/A	N/A
8000020	12/2010	Amitai et al.	N/A	N/A
8000491	12/2010	Brodkin et al.	N/A	N/A

8004765	12/2010	Amitai	N/A	N/A
8014050	12/2010	McGrew	N/A	N/A
8016475	12/2010	Travis	N/A	N/A
8018579	12/2010	Krah	N/A	N/A
8022942	12/2010	Bathiche et al.	N/A	N/A
8023783	12/2010	Mukawa et al.	N/A	N/A
RE42992	12/2010	David	N/A	N/A
8073296	12/2010	Mukawa et al.	N/A	N/A
8077274	12/2010	Sutherland et al.	N/A	N/A
8079713	12/2010	Ashkenazi	N/A	N/A
8082222	12/2010	Rangarajan et al.	N/A	N/A
8086030	12/2010	Gordon et al.	N/A	N/A
8089568	12/2011	Brown et al.	N/A	N/A
8093451	12/2011	Spangenberg et al.	N/A	N/A
8098439	12/2011	Amitai et al.	N/A	N/A
8105662	12/2011	Cherkaoui et al.	N/A	N/A
8107023	12/2011	Simmonds et al.	N/A	N/A
8107780	12/2011	Simmonds	N/A	N/A
8120548	12/2011	Barber	N/A	N/A
8120848	12/2011	Isano	N/A	N/A
8132948	12/2011	Owen et al.	N/A	N/A
8132976	12/2011	Odell et al.	N/A	N/A
8134434	12/2011	Diederichs et al.	N/A	N/A
8136690	12/2011	Fang et al.	N/A	N/A
8137981	12/2011	Andrew et al.	N/A	N/A
8142016	12/2011	Legerton et al.	N/A	N/A
8149086	12/2011	Klein et al.	N/A	N/A
8152315	12/2011	Travis et al.	N/A	N/A
8152353	12/2011	Yang et al.	N/A	N/A
8155489	12/2011	Saarikko et al.	N/A	N/A
8159752	12/2011	Wertheim et al.	N/A	N/A
8160409	12/2011	Large	N/A	N/A
8160411	12/2011	Levola et al.	N/A	N/A
D659137	12/2011	Matsumoto	N/A	N/A
8167173	12/2011	Simmonds et al.	N/A	N/A
8186874	12/2011	Sinbar et al.	N/A	N/A
8188925	12/2011	DeJean	N/A	N/A
8189263	12/2011	Wang et al.	N/A	N/A
8189973	12/2011	Travis et al.	N/A	N/A
D661334	12/2011	Cho et al.	N/A	N/A
D661335	12/2011	Jeon	N/A	N/A
8194325	12/2011	Levola et al.	N/A	N/A
8199803	12/2011	Hauske et al.	N/A	N/A
8202405	12/2011	Meneghini et al.	N/A	N/A
8213065	12/2011	Mukawa	N/A	N/A
8213755	12/2011	Mukawa et al.	N/A	N/A
8220966	12/2011	Mukawa	N/A	N/A
8224133	12/2011	Popovich et al.	N/A	N/A
8233204	12/2011	Robbins et al.	N/A	N/A
8253914	12/2011	Kajiya et al.	N/A	N/A

8254031	12/2011	Levola	N/A	N/A
8264498	12/2011	Vanderkamp et al.	N/A	N/A
8294749	12/2011	Cable	N/A	N/A
8295710	12/2011	Marcus	N/A	N/A
8301031	12/2011	Gentner et al.	N/A	N/A
8305577	12/2011	Kivioja et al.	N/A	N/A
8306423	12/2011	Gottwald et al.	N/A	N/A
8310327	12/2011	Willers et al.	N/A	N/A
8314819	12/2011	Kimmel et al.	N/A	N/A
8314993	12/2011	Levola et al.	N/A	N/A
8320032	12/2011	Levola	N/A	N/A
8321810	12/2011	Heintze	N/A	N/A
8325166	12/2011	Akutsu et al.	N/A	N/A
8329773	12/2011	Fäcke et al.	N/A	N/A
8335040	12/2011	Mukawa et al.	N/A	N/A
8335414	12/2011	Zinoviev et al.	N/A	N/A
D673996	12/2012	Kim et al.	N/A	N/A
8351744	12/2012	Travis et al.	N/A	N/A
8354640	12/2012	Hamre et al.	N/A	N/A
8354806	12/2012	Travis et al.	N/A	N/A
8355610	12/2012	Simmonds	N/A	N/A
8369019	12/2012	Baker et al.	N/A	N/A
8376548	12/2012	Schultz	N/A	N/A
8382293	12/2012	Phillips, III et al.	N/A	N/A
8384504	12/2012	Diederichs et al.	N/A	N/A
8384694	12/2012	Powell et al.	N/A	N/A
8384730	12/2012	Vanderkamp et al.	N/A	N/A
8396339	12/2012	Mukawa et al.	N/A	N/A
8396341	12/2012	Lee et al.	N/A	N/A
8398242	12/2012	Yamamoto et al.	N/A	N/A
8403490	12/2012	Sugiyama et al.	N/A	N/A
8422840	12/2012	Large	N/A	N/A
8427439	12/2012	Larsen et al.	N/A	N/A
8432363	12/2012	Saarikko et al.	N/A	N/A
8432372	12/2012	Butler et al.	N/A	N/A
8432614	12/2012	Amitai	N/A	N/A
8441731	12/2012	Sprague	N/A	N/A
8447365	12/2012	Immanuel	N/A	N/A
8466953	12/2012	Levola	N/A	N/A
8472119	12/2012	Kelly	N/A	N/A
8472120	12/2012	Border et al.	N/A	N/A
8477261	12/2012	Travis et al.	N/A	N/A
8481130	12/2012	Harding et al.	N/A	N/A
8482858	12/2012	Sprague	N/A	N/A
8488246	12/2012	Border et al.	N/A	N/A
8491121	12/2012	Tilleman et al.	N/A	N/A
8491136	12/2012	Travis et al.	N/A	N/A
8493366	12/2012	Bathiche et al.	N/A	N/A
8493562	12/2012	Kopp et al.	N/A	N/A
8493662	12/2012	Noui	N/A	N/A

8494229	12/2012	Jarvenpaa et al.	N/A	N/A
8503841	12/2012	Kopp et al.	N/A	N/A
8508848	12/2012	Saarikko	N/A	N/A
8520309	12/2012	Sprague	N/A	N/A
D691192	12/2012	Stanley et al.	N/A	N/A
8547638	12/2012	Levola	N/A	N/A
8548290	12/2012	Travers et al.	N/A	N/A
8565560	12/2012	Popovich et al.	N/A	N/A
D694310	12/2012	Cho et al.	N/A	N/A
D694311	12/2012	Cho et al.	N/A	N/A
8578038	12/2012	Kaikuranta et al.	N/A	N/A
8581831	12/2012	Travis	N/A	N/A
8582206	12/2012	Travis	N/A	N/A
8593734	12/2012	Laakkonen	N/A	N/A
8611014	12/2012	Valera et al.	N/A	N/A
8619062	12/2012	Powell et al.	N/A	N/A
D697130	12/2013	Lövgren	N/A	N/A
8633786	12/2013	Ermolov et al.	N/A	N/A
8634120	12/2013	Popovich et al.	N/A	N/A
8634139	12/2013	Brown et al.	N/A	N/A
8639072	12/2013	Popovich et al.	N/A	N/A
8643691	12/2013	Rosenfeld et al.	N/A	N/A
8643948	12/2013	Amitai et al.	N/A	N/A
8649099	12/2013	Schultz et al.	N/A	N/A
8654420	12/2013	Simmonds	N/A	N/A
8659826	12/2013	Brown et al.	N/A	N/A
D701206	12/2013	Luckey et al.	N/A	N/A
8670029	12/2013	McEldowney	N/A	N/A
8693087	12/2013	Nowatzyk et al.	N/A	N/A
8698705	12/2013	Burke	N/A	N/A
8731350	12/2013	Lin et al.	N/A	N/A
8736802	12/2013	Kajiya et al.	N/A	N/A
8736963	12/2013	Robbins et al.	N/A	N/A
8742952	12/2013	Bold	N/A	N/A
8746008	12/2013	Mauritsen et al.	N/A	N/A
8749886	12/2013	Gupta	N/A	N/A
8749890	12/2013	Wood et al.	N/A	N/A
8767294	12/2013	Chen et al.	N/A	N/A
8786923	12/2013	Chuang et al.	N/A	N/A
8810600	12/2013	Bohn et al.	N/A	N/A
8810913	12/2013	Simmonds et al.	N/A	N/A
8810914	12/2013	Amitai	N/A	N/A
8814691	12/2013	Haddick et al.	N/A	N/A
8816578	12/2013	Peng et al.	N/A	N/A
8817350	12/2013	Robbins et al.	N/A	N/A
8824836	12/2013	Sugiyama	N/A	N/A
8830143	12/2013	Pitchford et al.	N/A	N/A
8830584	12/2013	Saarikko et al.	N/A	N/A
8830588	12/2013	Brown et al.	N/A	N/A
8842368	12/2013	Simmonds et al.	N/A	N/A

8859412	12/2013	Jain	N/A	N/A
8872435	12/2013	Kreitzer et al.	N/A	N/A
8873149	12/2013	Bohn et al.	N/A	N/A
8873150	12/2013	Amitai	N/A	N/A
D718304	12/2013	Heinrich	N/A	N/A
D718366	12/2013	Mehin et al.	N/A	N/A
8885112	12/2013	Popovich et al.	N/A	N/A
8885997	12/2013	Nguyen et al.	N/A	N/A
8903207	12/2013	Brown et al.	N/A	N/A
8906088	12/2013	Pugh et al.	N/A	N/A
8913324	12/2013	Schrader	N/A	N/A
8913865	12/2013	Bennett	N/A	N/A
8917453	12/2013	Bohn	N/A	N/A
8917962	12/2013	Nichol et al.	N/A	N/A
8929589	12/2014	Publicover et al.	N/A	N/A
8933144	12/2014	Enomoto et al.	N/A	N/A
8934743	12/2014	Nishiwaki et al.	N/A	N/A
8937771	12/2014	Robbins et al.	N/A	N/A
8937772	12/2014	Burns et al.	N/A	N/A
8938141	12/2014	Magnusson	N/A	N/A
8950867	12/2014	Macnamara	N/A	N/A
8964298	12/2014	Haddick et al.	N/A	N/A
8965152	12/2014	Simmonds	N/A	N/A
D725102	12/2014	Lee et al.	N/A	N/A
8985803	12/2014	Bohn	N/A	N/A
8989535	12/2014	Robbins	N/A	N/A
D726180	12/2014	Roat et al.	N/A	N/A
9019595	12/2014	Jain	N/A	N/A
9025253	12/2014	Hadad et al.	N/A	N/A
9035344	12/2014	Jain	N/A	N/A
D733709	12/2014	Kawai	N/A	N/A
9075184	12/2014	Popovich et al.	N/A	N/A
9081178	12/2014	Simmonds et al.	N/A	N/A
9097890	12/2014	Miller et al.	N/A	N/A
9103978	12/2014	Nishiwaki et al.	N/A	N/A
9122015	12/2014	Shimizu	N/A	N/A
9128226	12/2014	Fattal et al.	N/A	N/A
9129295	12/2014	Border et al.	N/A	N/A
9164290	12/2014	Robbins et al.	N/A	N/A
9176324	12/2014	Scherer et al.	N/A	N/A
9188717	12/2014	Nishiwaki	N/A	N/A
9201270	12/2014	Fattal et al.	N/A	N/A
9215293	12/2014	Miller	N/A	N/A
D746896	12/2015	Markovitz et al.	N/A	N/A
9239507	12/2015	Chen et al.	N/A	N/A
9244275	12/2015	Li	N/A	N/A
9244280	12/2015	Tiana et al.	N/A	N/A
9244281	12/2015	Zimmerman et al.	N/A	N/A
D749074	12/2015	Cazalet et al.	N/A	N/A
9253359	12/2015	Takahashi	N/A	N/A

9269854	12/2015	Jain	N/A	N/A
D751551	12/2015	Ho et al.	N/A	N/A
D752129	12/2015	Lee et al.	N/A	N/A
9274338	12/2015	Robbins et al.	N/A	N/A
9274339	12/2015	Brown et al.	N/A	N/A
9274349	12/2015	Popovich et al.	N/A	N/A
D754782	12/2015	Kokinakis et al.	N/A	N/A
9310566	12/2015	Valera et al.	N/A	N/A
9316786	12/2015	Nishiwaki et al.	N/A	N/A
9329325	12/2015	Simmonds et al.	N/A	N/A
9335548	12/2015	Cakmakci et al.	N/A	N/A
9335604	12/2015	Popovich et al.	N/A	N/A
9341846	12/2015	Popovich et al.	N/A	N/A
9354366	12/2015	Jain	N/A	N/A
9366862	12/2015	Haddick et al.	N/A	N/A
9366864	12/2015	Brown et al.	N/A	N/A
9372347	12/2015	Levola et al.	N/A	N/A
9377623	12/2015	Robbins et al.	N/A	N/A
9377852	12/2015	Shapiro et al.	N/A	N/A
9389415	12/2015	Fattal et al.	N/A	N/A
9400395	12/2015	Travers et al.	N/A	N/A
9423360	12/2015	Kostamo et al.	N/A	N/A
9429692	12/2015	Saarikko et al.	N/A	N/A
9431794	12/2015	Jain	N/A	N/A
9435961	12/2015	Jiang	N/A	N/A
9456744	12/2015	Popovich et al.	N/A	N/A
9459451	12/2015	Saarikko et al.	N/A	N/A
9464779	12/2015	Popovich et al.	N/A	N/A
9465213	12/2015	Simmonds	N/A	N/A
9465227	12/2015	Popovich et al.	N/A	N/A
9484482	12/2015	Hsu et al.	N/A	N/A
9494799	12/2015	Robbins et al.	N/A	N/A
9507150	12/2015	Stratton et al.	N/A	N/A
9513480	12/2015	Saarikko et al.	N/A	N/A
9516193	12/2015	Aramaki	N/A	N/A
9519089	12/2015	Brown et al.	N/A	N/A
9519115	12/2015	Yashiki et al.	N/A	N/A
9523852	12/2015	Brown et al.	N/A	N/A
9535253	12/2016	Levola et al.	N/A	N/A
9541383	12/2016	Abovitz et al.	N/A	N/A
9541763	12/2016	Heberlein et al.	N/A	N/A
9547174	12/2016	Gao et al.	N/A	N/A
9551468	12/2016	Jones	N/A	N/A
9551874	12/2016	Amitai	N/A	N/A
9551880	12/2016	Amitai	N/A	N/A
9599813	12/2016	Stratton et al.	N/A	N/A
9612403	12/2016	Abovitz et al.	N/A	N/A
9632226	12/2016	Waldern et al.	N/A	N/A
9635352	12/2016	Henry et al.	N/A	N/A
9648313	12/2016	Henry et al.	N/A	N/A

9651368	12/2016	Abovitz et al.	N/A	N/A
9664824	12/2016	Simmonds et al.	N/A	N/A
9664910	12/2016	Mansharof et al.	N/A	N/A
9671612	12/2016	Kress et al.	N/A	N/A
9674413	12/2016	Tiana et al.	N/A	N/A
9678345	12/2016	Melzer et al.	N/A	N/A
9679367	12/2016	Wald	N/A	N/A
9715067	12/2016	Brown et al.	N/A	N/A
9715110	12/2016	Brown et al.	N/A	N/A
D793468	12/2016	Yu et al.	N/A	N/A
D795865	12/2016	Porter et al.	N/A	N/A
D795866	12/2016	Porter et al.	N/A	N/A
9726540	12/2016	Popovich et al.	N/A	N/A
9727772	12/2016	Popovich et al.	N/A	N/A
9733475	12/2016	Brown et al.	N/A	N/A
9739950	12/2016	Sqalli et al.	N/A	N/A
9746688	12/2016	Popovich et al.	N/A	N/A
9754507	12/2016	Wenger et al.	N/A	N/A
9762895	12/2016	Henry et al.	N/A	N/A
9766465	12/2016	Tiana et al.	N/A	N/A
9785231	12/2016	Zimmerman	N/A	N/A
9791694	12/2016	Haverkamp et al.	N/A	N/A
9791696	12/2016	Woltman et al.	N/A	N/A
9791703	12/2016	Vallius et al.	N/A	N/A
9804316	12/2016	Drolet et al.	N/A	N/A
9804389	12/2016	Popovich et al.	N/A	N/A
9823423	12/2016	Waldern et al.	N/A	N/A
9857605	12/2017	Popovich et al.	N/A	N/A
9874931	12/2017	Koenck et al.	N/A	N/A
9891436	12/2017	Wall et al.	N/A	N/A
9899800	12/2017	Ferrotti et al.	N/A	N/A
9915825	12/2017	Robbins et al.	N/A	N/A
9933684	12/2017	Brown et al.	N/A	N/A
9939577	12/2017	Inoue et al.	N/A	N/A
9939628	12/2017	Basset et al.	N/A	N/A
9959818	12/2017	Bohn	N/A	N/A
9977247	12/2017	Brown et al.	N/A	N/A
9989763	12/2017	Woltman et al.	N/A	N/A
10025093	12/2017	Wall et al.	N/A	N/A
D827641	12/2017	Morisawa	N/A	N/A
10067347	12/2017	Vallius et al.	N/A	N/A
10088675	12/2017	Brown et al.	N/A	N/A
10088686	12/2017	Robbins et al.	N/A	N/A
10089516	12/2017	Popovich et al.	N/A	N/A
10095045	12/2017	Robbins et al.	N/A	N/A
10107966	12/2017	Horibe et al.	N/A	N/A
10114220	12/2017	Grey et al.	N/A	N/A
10126552	12/2017	Brown et al.	N/A	N/A
10145533	12/2017	Popovich et al.	N/A	N/A
10156681	12/2017	Waldern et al.	N/A	N/A

10162181	12/2017	Webster et al.	N/A	N/A
10185154	12/2018	Popovich et al.	N/A	N/A
D840454	12/2018	Han et al.	N/A	N/A
10197804	12/2018	Stenberg et al.	N/A	N/A
10209517	12/2018	Popovich et al.	N/A	N/A
10216061	12/2018	Popovich et al.	N/A	N/A
10234696	12/2018	Popovich et al.	N/A	N/A
10241330	12/2018	Popovich et al.	N/A	N/A
10241332	12/2018	Vallius	N/A	N/A
10247943	12/2018	Yu et al.	N/A	N/A
10281725	12/2018	Yokoyama	N/A	N/A
10330777	12/2018	Popovich et al.	N/A	N/A
10345519	12/2018	Miller et al.	N/A	N/A
10359627	12/2018	Wall et al.	N/A	N/A
10359635	12/2018	Grey et al.	N/A	N/A
10359736	12/2018	Popovich et al.	N/A	N/A
D855687	12/2018	Villalpando	N/A	N/A
D859510	12/2018	Harmon et al.	N/A	N/A
10409144	12/2018	Popovich et al.	N/A	N/A
10423222	12/2018	Popovich et al.	N/A	N/A
10423813	12/2018	Popovich et al.	N/A	N/A
10437051	12/2018	Popovich et al.	N/A	N/A
10437064	12/2018	Popovich et al.	N/A	N/A
10444510	12/2018	Lee et al.	N/A	N/A
10459145	12/2018	Popovich et al.	N/A	N/A
10459311	12/2018	Popovich et al.	N/A	N/A
D871494	12/2018	Yamada et al.	N/A	N/A
10509241	12/2018	Robbins et al.	N/A	N/A
D872170	12/2019	Evans et al.	N/A	N/A
D872794	12/2019	Wilkins	N/A	N/A
10527797	12/2019	Waldern et al.	N/A	N/A
10532594	12/2019	Akahane et al.	N/A	N/A
10545346	12/2019	Waldern	N/A	G06F 3/0433
10551616	12/2019	Wall et al.	N/A	N/A
10560688	12/2019	Robbins	N/A	N/A
10569449	12/2019	Curts et al.	N/A	N/A
10578876	12/2019	Lam et al.	N/A	N/A
10591756	12/2019	Popovich et al.	N/A	N/A
10598938	12/2019	Huang et al.	N/A	N/A
D880575	12/2019	Thixton	N/A	N/A
10613268	12/2019	Colburn et al.	N/A	N/A
10642058	12/2019	Popovich et al.	N/A	N/A
10649119	12/2019	Mohanty et al.	N/A	N/A
10670808	12/2019	Trail	N/A	N/A
10670876	12/2019	Popovich et al.	N/A	N/A
10678053	12/2019	Waldern et al.	N/A	N/A
10690831	12/2019	Calafiore	N/A	N/A
10690851	12/2019	Waldern et al.	N/A	N/A
10690915	12/2019	Popovich et al.	N/A	N/A
10690916	12/2019	Popovich	N/A	G02F 1/292

10698214	12/2019	Vallius et al.	N/A	N/A
10705281	12/2019	Fattal et al.	N/A	N/A
10712571	12/2019	Popovich et al.	N/A	N/A
10725312	12/2019	Popovich et al.	N/A	N/A
10732351	12/2019	Colburn et al.	N/A	N/A
10732569	12/2019	Waldern et al.	N/A	N/A
10746989	12/2019	Brown et al.	N/A	N/A
10747982	12/2019	Popovich et al.	N/A	N/A
10795160	12/2019	Stanley et al.	N/A	N/A
10823887	12/2019	Calafiore et al.	N/A	N/A
10859768	12/2019	Popovich et al.	N/A	N/A
10859837	12/2019	Adema et al.	N/A	N/A
10890707	12/2020	Waldern et al.	N/A	N/A
10914950	12/2020	Waldern et al.	N/A	N/A
10942430	12/2020	Waldern et al.	N/A	N/A
10983257	12/2020	Colburn et al.	N/A	N/A
10983340	12/2020	Popovich et al.	N/A	N/A
10983346	12/2020	Vallius et al.	N/A	N/A
11009699	12/2020	Popovich et al.	N/A	N/A
11103892	12/2020	Liao et al.	N/A	N/A
11106048	12/2020	Popovich et al.	N/A	N/A
11107972	12/2020	Diest et al.	N/A	N/A
11137603	12/2020	Zhang	N/A	N/A
11169314	12/2020	Popovich et al.	N/A	N/A
11175512	12/2020	Waldern et al.	N/A	N/A
11194098	12/2020	Waldern et al.	N/A	N/A
11194159	12/2020	Popovich et al.	N/A	N/A
11194162	12/2020	Waldern et al.	N/A	N/A
11204540	12/2020	Popovich et al.	N/A	N/A
11231544	12/2021	Lin et al.	N/A	N/A
11243333	12/2021	Ouderkirk et al.	N/A	N/A
11256155	12/2021	Popovich et al.	N/A	N/A
11281013	12/2021	Popovich	N/A	G02B 27/0179
11287666	12/2021	Popovich et al.	N/A	N/A
11300795	12/2021	Stanley et al.	N/A	N/A
11306193	12/2021	Lane et al.	N/A	N/A
11307357	12/2021	Mohanty	N/A	N/A
11307432	12/2021	Popovich et al.	N/A	N/A
11340386	12/2021	Ouderkirk et al.	N/A	N/A
11378732	12/2021	Waldern et al.	N/A	N/A
11391950	12/2021	Calafiore	N/A	N/A
11402801	12/2021	Waldern et al.	N/A	N/A
11442222	12/2021	Waldern et al.	N/A	N/A
11448937	12/2021	Brown et al.	N/A	N/A
11460621	12/2021	Popovich et al.	N/A	N/A
11480788	12/2021	Popovich et al.	N/A	N/A
11487131	12/2021	Popovich et al.	N/A	N/A
11513350	12/2021	Waldern et al.	N/A	N/A
11543594	12/2022	Grant et al.	N/A	N/A

11561409	12/2022	Popovich et al.	N/A	N/A
11573483	12/2022	Waldern et al.	N/A	N/A
11579455	12/2022	Stanley et al.	N/A	N/A
11586046	12/2022	Waldern et al.	N/A	N/A
11592614	12/2022	Waldern et al.	N/A	N/A
11604314	12/2022	Popovich et al.	N/A	N/A
11703799	12/2022	Waldern et al.	N/A	N/A
11709373	12/2022	Popovich et al.	N/A	N/A
11726323	12/2022	Popovich et al.	N/A	N/A
11726329	12/2022	Popovich et al.	N/A	N/A
11740472	12/2022	Popovich et al.	N/A	N/A
11747568	12/2022	Waldern et al.	N/A	N/A
11747719	12/2022	Popovich et al.	N/A	N/A
11754842	12/2022	Popovich	345/8	G02B 27/0093
11815781	12/2022	Brown et al.	N/A	N/A
11874477	12/2023	Popovich et al.	N/A	N/A
11899238	12/2023	Waldern et al.	N/A	N/A
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2003/0067685	12/2002	Niv	N/A	N/A
2003/0076590	12/2002	Kramer	N/A	N/A
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2009/0052046	12/2008	Amitai	N/A	N/A
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2009/0128781	12/2008	Li	N/A	N/A
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2009/0128911	12/2008	Itzkovitch et al.	N/A	N/A
2009/0136246	12/2008	Murakami	N/A	N/A
2009/0141324	12/2008	Mukawa	N/A	N/A
2009/0153437	12/2008	Aharoni	N/A	N/A
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2010/0165465	12/2009	Levola	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/655,756 entitled Apparatus for Providing Waveguide Displays with Two-Dimensional Pupil Expansion filed on Mar. 21, 2022, which application is a continuation of U.S. patent application Ser. No. 17/118,285 entitled Apparatus for Providing Waveguide Displays with Two-Dimensional Pupil Expansion filed on Dec. 10, 2020, which application is a continuation of U.S. patent application Ser. No. 16/806,947 entitled Apparatus for Providing Waveguide Displays with Two-Dimensional Pupil Expansion filed on Mar. 2, 2020, which application is a continuation of U.S. patent application Ser. No. 15/765,243 entitled Apparatus for Providing Waveguide Displays with Two-Dimensional Pupil Expansion filed on Mar. 30, 2018, which application is a national stage of PCT Application No. PCT/GB2016/00018 entitled Waveguide Display filed on Oct. 4, 2016, which application claims priority to U.S. Provisional Patent Application No. 62/284,603 entitled Waveguide Display filed on Oct. 5, 2015 and U.S. Provisional Patent Application No. 62/285,275 entitled Waveguide Displays filed on Oct. 23, 2015, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND OF THE INVENTION

(1) The present disclosure relates to displays including but not limited to near eye displays and more particularly to holographic waveguide displays.

(2) Waveguide optics is currently being considered for a range of display and sensor applications for which the ability of waveguides to integrate multiple optical functions into a thin, transparent, lightweight substrate is of key importance. This new approach is stimulating new product developments including near-eye displays for Augmented Reality (AR) and Virtual Reality (VR),

compact Heads Up Display (HUDs) for aviation and road transport and sensors for biometric and laser radar (LIDAR) applications. Waveguide displays have been proposed which use diffraction gratings to preserve eye box size while reducing lens size. U.S. Pat. No. 4,309,070 issued to St. Leger Searle and U.S. Pat. No. 4,711,512 issued to Upatnieks disclose substrate waveguide head up displays where the pupil of a collimating optical system is effectively expanded by the waveguide structure. U.S. patent application Ser. No. 13/869,866 discloses holographic wide angle displays and U.S. patent application Ser. No. 13/844,456 discloses waveguide displays having an upper and lower field of view.

(3) A common requirement in waveguide optics is to provide beam expansion in two orthogonal directions. In display applications this translates to a large eyebox. While the principles of beam expansion in holographic waveguides are well established dual axis expansion requires separate grating layers to provide separate vertical and horizontal expansion. One of the gratings, usually the one giving the second axis expansion, also provides the near eye component of the display where the high transparency and thin, lightweight form factor of a diffractive optics can be used to maximum effect. In practical display applications, which demand full color and large fields of view the number of layers required to implement dual axis expansion becomes unacceptably large resulting in increased thickness weight and haze. Solutions for reducing the number of layers based on multiplexing two or more gratings in a single layer or fold gratings which can perform dual axis expansion (for a given angular range and wavelength) in a single layer are currently in development. Dual axis expansion is also an issue in waveguides for sensor applications such as eye trackers and LIDAR. There is a requirement for a low cost, efficient, compact dual axis expansion waveguide.

SUMMARY OF THE INVENTION

(4) It is a first object of the invention to provide a low cost, efficient, compact dual axis expansion waveguide.

(5) The object of the invention is achieved in first embodiment of the invention in which there is provided an optical display, comprising: a first waveguide comprising a first surface and a second surface, an input coupler, a fold grating, and an output grating. The input coupler is configured to receive collimated first wavelength light from an Input Image Node (IIN) and to cause the light to travel within the first waveguide via total internal reflection between the first surface and the second surface to the fold grating. The fold grating is configured to provide pupil expansion in a first direction and to direct the light to the output grating via total internal reflection between the first surface and the second surface. The output grating is configured to provide pupil expansion in a second direction different than the first direction and to cause the light to exit the first waveguide from the first surface or the second surface. At least one of the input coupler, the fold grating or the output grating is a rolled k-vector grating. The light undergoes a dual interaction with the fold grating.

(6) In some embodiments the IIN comprises a light source, a microdisplay for displaying image pixels and collimation optics. The IIN projects the image displayed on the microdisplay panel such that each image pixel is converted into a unique angular direction within the first waveguide

(7) In some embodiments at least one of the gratings is switchable between a diffracting and non-diffracting state.

(8) In some embodiments the optical display further comprises a second waveguide comprising a first surface and a second surface, an input coupler, a fold grating, and an output grating, wherein the input coupler is configured to receive collimated second wavelength light from the IIN.

(9) In some embodiments at least one of the input coupler, the fold grating, and the output grating is a liquid crystal-based grating.

(10) In some embodiments the first direction is orthogonal to the second direction.

(11) In some embodiments the first direction is horizontal and the second direction is vertical.

(12) In some embodiments the optical display further comprises an eye tracker.

- (13) In some embodiments the optical display further comprises a dynamic focus lens disposed in the IIN.
- (14) In some embodiments the optical display further comprises a dynamic focus lens disposed in proximity to the first or second surface of the first waveguide.
- (15) In some embodiments the first waveguide further comprises a first optical interface the IIN further comprises a second optical interface wherein the first and second optical interface can be decoupled by one of a mechanical mechanism or a magnetic mechanism.
- (16) In some embodiments the first waveguide is disposable. In some embodiments the first surface and the second surface are planar surfaces. In some embodiments the first surface and the second surface are curved surfaces. In some embodiments the IIN comprises a laser scanner.
- (17) In some embodiments the display provides one of a HMD, a HUD, an eye-slaved display, a dynamic focus display or a light field display.
- (18) In some embodiments at least one of the input coupler, fold grating and output grating multiplexes at least one of color or angle.
- (19) In some embodiments the optical display further comprises a beam homogenizer.
- (20) In some embodiments the display includes at least one optical traversing a gradient index image transfer waveguide.
- (21) In some embodiments the optical display further comprises a dichroic filter disposed between the input grating regions of the first and second waveguides.
- (22) In some embodiments the IIN further comprises a spatially-varying numerical aperture component for providing a numerical aperture variation along a direction corresponding to the field of view coordinate diffracted by the input coupler.
- (23) In some embodiments the spatially-varying numerical aperture component has at least one of diffractive, birefringent, refracting or scattering characteristics.
- (24) In some embodiments the field of view coordinate is the horizontal field of view of the display.
- (25) In some embodiments a spatially varying-numerical aperture is provided by tilting a stop plane such that its normal vector is aligned perpendicular to the highest display field angle in the plane containing the field of view coordinate diffracted by the input coupler.
- (26) One exemplary embodiment of the disclosure relates to a near eye optical display. The near eye optical display includes a waveguide comprising a first surface and a second surface, an input coupler, a fold grating, and an output grating. The input coupler is configured to receive collimated light from a display source and to cause the light to travel within the waveguide via total internal reflection between the first surface and the second surface to the fold grating. The fold grating is configured to provide pupil expansion in a first direction and to direct the light to the output grating via total internal reflection between the first surface and the second surface. The output grating is configured to provide pupil expansion in a second direction different than the first direction and to cause the light to exit the waveguide from the first surface or the second surface.
- (27) Another exemplary embodiment of the disclosure relates to a method of displaying information. The method includes receiving collimated light in a waveguide having a first surface and a second surface; providing the collimated light to a fold grating via total internal reflection between the first surface and the second surface; providing pupil expansion in a first direction using the fold grating and directing the light to an output grating via total internal reflection between the first surface and the second surface; and providing pupil expansion in a second direction different than the first direction and causing the light to exit the waveguide from the first surface or the second surface.
- (28) Following below are more detailed descriptions of various concepts related to, and embodiments of, an inventive optical display and methods for displaying information. It should be appreciated that various concepts introduced above and discussed in greater detail below may be implemented in any of numerous ways, as the disclosed concepts are not limited to any particular

manner of implementation. Examples of specific implementations and applications are provided primarily for illustrative purposes. A more complete understanding of the invention can be obtained by considering the following detailed description in conjunction with the accompanying drawings, wherein like index numerals indicate like parts. For purposes of clarity, details relating to technical material that is known in the technical fields related to the invention have not been described in detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross section view of a waveguide display in one embodiment.
- (2) FIG. 2 is a schematic plan view of a waveguide display shown the disposition of the gratings in one grating layer in one embodiment.
- (3) FIG. 3 is a schematic plan view of a waveguide display shown the disposition of the gratings in two grating layers in one embodiment.
- (4) FIG. 4 is a schematic cross section view of a color waveguide display using one waveguide per color and two grating layers in each waveguide in one embodiment.
- (5) FIG. 5 is a schematic cross section view of a color waveguide display using one waveguide per color and one grating layer in each waveguide in one embodiment.
- (6) FIG. 6 is a schematic plan view of a waveguide display showing an input grating a fold grating and an output grating in one embodiment.
- (7) FIG. 7 is a cross section view of an eye tracked near eye display according to the principles of the invention in one embodiment.
- (8) FIG. 8 is a cross section view of an eye tracked near eye display incorporating a dynamic focus lens in one embodiment.
- (9) FIG. 9 is a cross section view of an eye tracked near eye display incorporating a dynamic focus lens in one embodiment.
- (10) FIG. 10A is a schematic plan view of a first operational state display comprising a waveguide that can be decoupled from the IIN in one embodiment.
- (11) FIG. 10B is a schematic plan view of a second operational state display comprising a waveguide that can be decoupled from the IIN in one embodiment.
- (12) FIG. 11A is a rolled K-vector grating providing stepwise changes in K-vector direction in one embodiment.
- (13) FIG. 11B is a rolled K-vector grating providing continuous changes in K-vector direction in one embodiment.
- (14) FIG. 12A is front view of a waveguide display eyepiece in one embodiment.
- (15) FIG. 12B is plan view of a waveguide display eyepiece in one embodiment.
- (16) FIG. 12C is side view of a waveguide display eyepiece in one embodiment.
- (17) FIG. 12D is a three dimensional view of a waveguide display eyepiece in one embodiment.
- (18) FIG. 13A is a three dimensional view of a waveguide display implemented in a motorcycle helmet in one embodiment.
- (19) FIG. 13B is another three dimensional view of a waveguide display implemented in a motorcycle helmet in one embodiment.
- (20) FIG. 14 is a schematic cross section view of a reflective microdisplay input image node containing a spatially-varying numerical aperture component in one embodiment.
- (21) FIG. 15 is a schematic cross section view of a reflective microdisplay input image node containing a spatially-varying numerical aperture component in one embodiment.
- (22) FIG. 16 is a schematic cross section view of a transmissive microdisplay input image node containing a spatially-varying numerical aperture component in one embodiment.

- (23) FIG. 17 is a schematic cross section view of an emissive microdisplay input image node containing a spatially-varying numerical aperture component in one embodiment.
- (24) FIG. 18A is a schematic cross section view of a spatially-varying numerical aperture component based on a wedge prism in one embodiment.
- (25) FIG. 18B is a schematic cross section view of a spatially-varying numerical aperture component based on a wedge prism with one curved surface in one embodiment.
- (26) FIG. 18C is a schematic cross section view of a spatially-varying numerical aperture component based on an array of prisms in one embodiment.
- (27) FIG. 18D is a schematic cross section view of a spatially-varying numerical aperture component based on an array of lenses in one embodiment.
- (28) FIG. 19A is a schematic cross section view of a spatially-varying numerical aperture component based on an array of scattering elements in one embodiment.
- (29) FIG. 19B is a schematic cross section view of a spatially-varying numerical aperture component based on a substrate with a continuously varying scattering function in one embodiment.
- (30) FIG. 19C is a schematic cross section view of a spatially-varying numerical aperture component based on a substrate with a continuously varying birefringence tensor in one embodiment.
- (31) FIG. 19D is a schematic cross section view of a spatially-varying numerical aperture component based on an array of grating elements in one embodiment.
- (32) FIG. 20 is a schematic cross section view of an optical arrangement for providing varying numerical aperture across a pupil using a tilted pupil plane in one embodiment.
- (33) FIG. 21A is a three dimensional view of a first operational state of a wearable display comprising a retractable waveguide in one embodiment.
- (34) FIG. 21B is a three dimensional view of a second operational state of a wearable display comprising a retractable waveguide in one embodiment.
- (35) FIG. 21C is a three dimensional view of a third operational state of a wearable display comprising a retractable waveguide in one embodiment.
- (36) FIG. 22A is front view of a waveguide component showing the input, fold and output gratings in one embodiment.
- (37) FIG. 22B is front view of a waveguide component showing the input, fold and output gratings in one embodiment.
- (38) FIG. 23 is front view of a waveguide component showing the input, fold and output gratings in one embodiment.
- (39) FIG. 24 is a three dimensional view of a near eye display showing a ray trace from the IIN and waveguide component up to the eye box in one embodiment.
- (40) FIG. 25A is a three dimensional view of a first operational state of a motorcycle display in one embodiment.
- (41) FIG. 25B is a three dimensional view of a second operational state of a motorcycle display in one embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(42) The invention will now be further described by way of example only with reference to the accompanying drawings. It will be apparent to those skilled in the art that the present invention may be practiced with some or all of the present invention as disclosed in the following description. For the purposes of explaining the invention well-known features of optical technology known to those skilled in the art of optical design and visual displays have been omitted or simplified in order not to obscure the basic principles of the invention. Unless otherwise stated the term “on-axis” in relation to a ray or a beam direction refers to propagation parallel to an axis normal to the surfaces of the optical components described in relation to the invention. In the following description the terms light, ray, beam and direction may be used interchangeably and in association with each other

to indicate the direction of propagation of electromagnetic radiation along rectilinear trajectories. The term light and illumination may be used in relation to the visible and infrared bands of the electromagnetic spectrum. Parts of the following description will be presented using terminology commonly employed by those skilled in the art of optical design. It should also be noted that in the following description of the invention repeated usage of the phrase “in one embodiment” does not necessarily refer to the same embodiment.

(43) Referring generally to the Figures, systems and methods relating to near-eye display or head up display systems are shown according to various embodiments. Holographic waveguide technology can be advantageously utilized in waveguides for helmet mounted displays or head mounted displays (HMDs) and head up displays (HUDs) for many applications, including military applications and consumer applications (e.g., augmented reality glasses, etc.). Switchable Bragg gratings (SBGs) may be used in waveguides to eliminate extra layers and to reduce the thickness of current display systems, including HMDs, HUDs, and other near eye displays and to increase the field of view by tiling images presented sequentially on a microdisplay. A larger exit pupil may be created by using fold gratings in conjunction with conventional gratings to provide pupil expansion on a single waveguide in both the horizontal and vertical directions. Using the systems and methods disclosed herein, a single optical waveguide substrate may generate a wider field of view than found in current waveguide systems. Diffraction gratings may be used to split and diffract light rays into several beams that travel in different directions, thereby dispersing the light rays.

(44) The grating used in the invention is desirably a Bragg grating (also referred to as a volume grating). Bragg gratings have high efficiency with little light being diffracted into higher orders. The relative amount of light in the diffracted and zero order can be varied by controlling their refractive index modulation of the grating, a property which is used to make lossy waveguide gratings for extracting light over a large pupil. One important class of gratings is known as Switchable Bragg Gratings (SBG). SBGs are fabricated by first placing a thin film of a mixture of photopolymerizable monomers and liquid crystal material between parallel glass plates. One or both glass plates support electrodes, typically transparent indium tin oxide films, for applying an electric field across the film. A volume phase grating is then recorded by illuminating the liquid material (often referred to as the syrup) with two mutually coherent laser beams, which interfere to form a slanted fringe grating structure. During the recording process, the monomers polymerize and the mixture undergoes a phase separation, creating regions densely populated by liquid crystal micro-droplets, interspersed with regions of clear polymer. The alternating liquid crystal-rich and liquid crystal-depleted regions form the fringe planes of the grating. The resulting volume phase grating can exhibit very high diffraction efficiency, which may be controlled by the magnitude of the electric field applied across the film. When an electric field is applied to the grating via transparent electrodes, the natural orientation of the LC droplets is changed causing the refractive index modulation of the fringes to reduce and the hologram diffraction efficiency to drop to very low levels. Typically, SBG Elements are switched clear in 30 μ s. With a longer relaxation time to switch ON. Note that the diffraction efficiency of the device can be adjusted, by means of the applied voltage, over a continuous range. The device exhibits near 100% efficiency with no voltage applied and essentially zero efficiency with a sufficiently high voltage applied. In certain types of HPDLC devices magnetic fields may be used to control the LC orientation. In certain types of HPDLC phase separation of the LC material from the polymer may be accomplished to such a degree that no discernible droplet structure results. A SBG may also be used as a passive grating. In this mode its chief benefit is a uniquely high refractive index modulation.

(45) SBGs may be used to provide transmission or reflection gratings for free space applications. SBGs may be implemented as waveguide devices in which the HPDLC forms either the waveguide core or an evanescently coupled layer in proximity to the waveguide. The parallel glass plates used to form the HPDLC cell provide a total internal reflection (TIR) light guiding structure. Light is coupled out of the SBG when the switchable grating diffracts the light at an angle beyond the TIR

condition. Waveguides are currently of interest in a range of display and sensor applications. Although much of the earlier work on HPDLC has been directed at reflection holograms transmission devices are proving to be much more versatile as optical system building blocks. Typically, the HPDLC used in SBGs comprise liquid crystal (LC), monomers, photoinitiator dyes, and coinitiators. The mixture frequently includes a surfactant. The patent and scientific literature contains many examples of material systems and processes that may be used to fabricate SBGs. Two fundamental patents are: U.S. Pat. No. 5,942,157 by Sutherland, and U.S. Pat. No. 5,751,452 by Tanaka et al. Both filings describe monomer and liquid crystal material combinations suitable for fabricating SBG devices. One of the known attributes of transmission SBGs is that the LC molecules tend to align normal to the grating fringe planes. The effect of the LC molecule alignment is that transmission SBGs efficiently diffract P polarized light (ie light with the polarization vector in the plane of incidence) but have nearly zero diffraction efficiency for S polarized light (ie light with the polarization vector normal to the plane of incidence. Transmission SBGs may not be used at near-grazing incidence as the diffraction efficiency of any grating for P polarization falls to zero when the included angle between the incident and reflected light is small.

(46) The object of the invention is achieved in first embodiment illustrated in FIG. 1 in which there is provided a dual axis expansion waveguide display **100** comprising a light source **101** a microdisplay panel **102** and an input image node (IIN) **103** optically coupled to a waveguide **104** comprise two grating layers **104A,104B**. In some embodiments the waveguide is formed by sandwiched the grating layers between glass or plastic substrates to form a stack within which total internal reflection occurs at the outer substrate and air interfaces. The stack may further comprise additional layers such as beam splitting coatings and environmental protection layers. Each grating layer contains an input grating **105A,105B**, a fold grating exit pupil expander **106A,106B** and an output grating **107A,107B** where characters A and B refer to waveguide layers **104A,104B** respectively. The input grating, fold grating and the output grating are holographic gratings, such as a switchable or non-switchable SBG. As used herein, the term grating may encompass a grating comprised of a set of gratings in some embodiments. In general the IIN integrates a microdisplay panel, light source and optical components needed to illuminate the display panel, separate the reflected light and collimate it into the required FOV. In the embodiment of FIG. 1 and in the embodiments to be described below at least one of the input fold and output gratings may be electrically switchable. In many embodiments it is desirable that all three grating types are passive, that is, non-switching. The IIN projects the image displayed on the microdisplay panel such that each display pixel is converted into a unique angular direction within the substrate waveguide according to some embodiments. The collimation optics contained in the IIN may comprise lens and mirrors which in some embodiments may be diffractive lenses and mirrors.

(47) In some embodiments the IIN may be based on the embodiments and teachings disclosed in U.S. patent application Ser. No.: 13/869,866 entitled HOLOGRAPHIC WIDE ANGLE DISPLAY, and U.S. patent application Ser. No.: 13/844,456 entitled TRANSPARENT WAVEGUIDE DISPLAY. In some embodiments the IIN contains beamsplitter for directing light onto the microdisplay and transmitting the reflected light towards the waveguide. In one embodiment the beamsplitter is a grating recorded in HPDLC and uses the intrinsic polarization selectivity of such gratings to separate the light illuminating the display and the image modulated light reflected off the display. In some embodiments the beam splitter is a polarizing beam splitter cube. In some embodiment the IIN incorporates a despeckler. Advantageously, the despeckler is holographic waveguide device based on the embodiments and teachings of U.S. Pat. No. 8,565,560 entitled LASER ILLUMINATION DEVICE.

(48) The light source can be a laser or LED and can include one or more lenses for modifying the illumination beam angular characteristics. The image source can be a micro-display or laser based display. LED will provide better uniformity than laser. If laser illumination is used there is a risk of illumination banding occurring at the waveguide output. In some embodiments laser illumination

banding in waveguides can be overcome using the techniques and teachings disclosed in U.S. Provisional Patent Application No.: 62/071,277 entitled METHOD AND APPARATUS FOR GENERATING INPUT IMAGES FOR HOLOGRAPHIC WAVEGUIDE DISPLAYS. In some embodiments, the light from the light source **101** is polarized. In one or more embodiments, the image source is a liquid crystal display (LCD) micro display or liquid crystal on silicon (LCoS) micro display.

(49) The light path from the source to the waveguide via the IIN is indicated by rays **1000-1003**. The input grating of each grating layer couples a portion of the light into a TIR path in the waveguide once such path being represented by the rays **1004-1005**. The output waveguides **107A,107C** diffract light out of the waveguide into angular ranges of collimated light **1006,1007** respectively for viewing by the eye **108**. The angular ranges, which correspond to the field of view of the display, are defined solely by the IIN optics. In some embodiments the waveguide gratings may encoded optical power for adjusting the collimation of the output. In some embodiments the output image is at infinity. In some embodiments the output image may be formed at distances of several meters from the eye box. Typically the eye is positioned within the exit pupil or eye box of the display.

(50) In some embodiments similar to the one shown in FIG. **1** each grating layer addresses half the total field of view. Typically, the fold gratings are clocked (that is, tilted in the waveguide plane) at 45° to ensure adequate angular bandwidth for the folded light. However some embodiments of the invention may use other clock angles to satisfy spatial constraints on the positioning of the gratings that may arise in the ergonomic design of the display. In some embodiments at least one of the input and output gratings have rolled k-vectors. The K-vector is a vector aligned normal to the grating planes (or fringes) which determines the optical efficiency for a given range of input and diffracted angles. Rolling the K-vectors allows the angular bandwidth of the grating to be expanded without the need to increase the waveguide thickness.

(51) In some embodiments the fold grating angular bandwidth can be enhanced by designing the grating prescription provides dual interaction of the guided light with the grating. Exemplary embodiments of dual interaction fold gratings are disclosed in U.S. patent application Ser. No.: 14/620,969 entitled WAVEGUIDE GRATING DEVICE.

(52) It is well established in the literature of holography that more than one holographic prescription can be recorded into a single holographic layer. Methods for recording such multiplexed holograms are well known to those skilled in the art. In some embodiments at least one of the input, fold or output gratings may combine two or more angular diffraction prescriptions to expand the angular bandwidth. Similarly, in some embodiments at least one of the input, fold or output gratings may combine two or more spectral diffraction prescriptions to expand the spectral bandwidth. For example a color multiplexed grating may be used to diffract two or more of the primary colors.

(53) FIG. **2** is a plan view **110** of a single grating layer similar to the ones used in FIG. **1**. The grating layer **111** which is optically coupled to the IIN **112** comprising input grating **113**, a first beamsplitter **114**, a fold grating **115**, a second beamsplitter **116** and an output grating **117**. The beamsplitter are partially transmitting coatings which homogenise the wave guided light by providing multiple reflection paths within the waveguide. Each beamsplitter may comprise more than one coating layer with each coating layer being applied to a transparent substrates. Typical beam paths from the IIN up to the eye **118** are indicated by the rays **1010-1014**.

(54) FIG. **3** is a plan view **110** of a two grating layer configuration similar to the ones used in FIG. **1** The grating layers **121A,121B** which are optically coupled to the IIN **122** comprise input gratings **123A,123B**, first beamsplitters **124A,124B**, fold gratings **125A,125B**, second beamsplitters **126A,126B** and output gratings **127A,127B**, where the characters A,B refer to the first and second grating layers and the gratings and beams splitters of the two layers substantially overlap.

(55) In the most waveguide configurations the input fold and output gratings are formed in a single

layer sandwiched by transparent substrates. The embodiment of FIG. 1 has two such layers stacked. In some embodiments the waveguide may comprise just one grating layer. The substrates are not illustrated in FIG. 1 where the gratings are switching transparent electrodes are applied to opposing surfaces of the substrate layers sandwiching the switching grating. In some embodiments the cell substrates may be fabricated from glass. An exemplary glass substrate is standard Corning Willow glass substrate (index 1.51) which is available in thicknesses down to 50 micron. In other embodiments the cell substrates may be optical plastics.

(56) In some embodiments the grating layer may be broken up into separate layers. For example, in some embodiments, a first layer includes the fold grating while a second layer includes the output grating. In some embodiments, a third layer can include the input grating. The number of layers may then be laminated together into a single waveguide substrate. In some embodiments, the grating layer is comprised of a number of pieces including the input coupler, the fold grating and the output grating (or portions thereof) that are laminated together to form a single substrate waveguide. The pieces may be separated by optical glue or other transparent material of refractive index matching that of the pieces. In another embodiment, the grating layer may be formed via a cell making process by creating cells of the desired grating thickness and vacuum filling each cell with SBG material for each of the input coupler, the fold grating and the output grating. In one embodiment, the cell is formed by positioning multiple plates of glass with gaps between the plates of glass that define the desired grating thickness for the input coupler, the fold grating and the output grating. In one embodiment, one cell may be made with multiple apertures such that the separate apertures are filled with different pockets of SBG material. Any intervening spaces may then be separated by a separating material (e.g., glue, oil, etc.) to define separate areas. In one embodiment the SBG material may be spin-coated onto a substrate and then covered by a second substrate after curing of the material. By using the fold grating, the waveguide display advantageously requires fewer layers than previous systems and methods of displaying information according to some embodiments. In addition, by using fold grating, light can travel by total internal reflection within the waveguide in a single rectangular prism defined by the waveguide outer surfaces while achieving dual pupil expansion. In another embodiment, the input coupler, the fold grating and the output grating can be created by interfering two waves of light at an angle within the substrate to create a holographic wave front, thereby creating light and dark fringes that are set in the waveguide substrate **101** at a desired angle. In some embodiments the grating in a given layer is recorded in stepwise fashion by scanning or stepping the recording laser beams across the grating area. In some embodiments the gratings are recorded using mastering and contact copying process currently used in the holographic printing industry.

(57) In one embodiment, the input coupler, the fold grating, and the output grating embodied as SBGs can be Bragg gratings recorded in a holographic polymer dispersed liquid crystal (HPDLC) (e.g., a matrix of liquid crystal droplets), although SBGs may also be recorded in other materials. In one embodiment, SBGs are recorded in a uniform modulation material, such as POLICRYPS or POLIPHEN having a matrix of solid liquid crystals dispersed in a liquid polymer. The SBGs can be switching or non-switching in nature. In its non switching form an SBG has the advantage over conventional holographic photopolymer materials of being capable of providing high refractive index modulation due to its liquid crystal component. Exemplary uniform modulation liquid crystal-polymer material systems are disclosed in United State Patent Application Publication No.: US2007/0019152 by Caputo et al and PCT Application No.: PCT/EP2005/006950 by Stumpe et al. both of which are incorporated herein by reference in their entireties. Uniform modulation gratings are characterized by high refractive index modulation (and hence high diffraction efficiency) and low scatter.

(58) In one embodiment, the input coupler, the fold grating, and the output grating a reverse mode HPDLC material. Reverse mode HPDLC differs from conventional HPDLC in that the grating is passive when no electric field is applied and becomes diffractive in the presence of an electric field.

The reverse mode HPDLC may be based on any of the recipes and processes disclosed in PCT Application No.: PCT/GB2012/000680, entitled IMPROVEMENTS TO HOLOGRAPHIC POLYMER DISPERSED LIQUID CRYSTAL MATERIALS AND DEVICES. The grating may be recorded in any of the above material systems but used in a passive (non-switching) mode. The fabrication process is identical to that used for switched but with the electrode coating stage being omitted. LC polymer material systems are highly desirable in view of their high index modulation. In some embodiments the gratings are recorded in HPDLC but are not switched.

(59) In some embodiments the input grating may be replaced by another type of input coupler such as a prism, or reflective surface. In some embodiments, the input coupler can be a holographic grating, such as a switchable or non-switchable SBG grating. The input coupler is configured to receive collimated light from a display source and to cause the light to travel within the waveguide via total internal reflection between the first surface and the second surface to the fold grating. The input coupler may be orientated directly towards or at an angle relative to the fold grating. For example, in one embodiment, the input coupler may be set at a slight incline in relation to the fold grating.

(60) In some embodiments, the fold grating may be oriented in a diagonal direction. The fold grating is configured to provide pupil expansion in a first direction and to direct the light to the output grating via total internal reflection inside the waveguide in some embodiments.

(61) In one embodiment, a longitudinal edge of each fold grating is oblique to the axis of alignment of the input coupler such that each fold grating is set on a diagonal with respect to the direction of propagation of the display light. The fold grating is angled such that light from the input coupler is redirected to the output grating. In one example, the fold grating is set at a forty-five degree angle relative to the direction that the display image is released from the input coupler. This feature causes the display image propagating down the fold grating to be turned into the output grating. For example, in one embodiment, the fold grating causes the image to be turned 90 degrees into the output grating. In this manner, a single waveguide provides dual axis pupil expansion in both the horizontal and vertical directions. In one embodiment, each of the fold grating may have a partially diffractive structure.

(62) In some embodiments, each of the fold gratings may have a fully diffractive structure. In some embodiments, different grating configurations and technologies may be incorporated in a single waveguide.

(63) The output grating is configured to provide pupil expansion in a second direction different than the first direction and to cause the light to exit the waveguide from the first surface or the second surface. The output grating receives the display image from the fold grating via total internal reflection and provides pupil expansion in a second direction. In some embodiments, the output grating consists of multiple layers of substrate, thereby comprising multiple layers of output gratings. Accordingly, there is no requirement for gratings to be in one plane within the waveguide, and gratings may be stacked on top of each other (e.g., cells of gratings stacked on top of each other).

(64) In some embodiments, a quarter wave plate on the substrate waveguide rotates polarization of a light ray to maintain efficient coupling with the SBGs. The quarter wave plate may be coupled to or adhered to the surface of substrate waveguide **101**. For example, in one embodiment, the quarter wave plate is a coating that is applied to substrate waveguide. The quarter wave plate provides light wave polarization management. Such polarization management may help light rays retain alignment with the intended viewing axis by compensating for skew waves in the waveguide. The quarter wave plate is optional and can increase the efficiency of the optical design in some embodiments. In some embodiments, the waveguide does not include the quarter wave plate **142**. The quarter wave plate may be provided as multi-layer coating.

(65) The embodiment of FIG. **1** would normally be operated in monochrome. A color would comprises a stack of monochrome waveguides of similar design to the one in FIG. **1**. The design

may use red green and blue waveguide layers as shown or, alternatively, red and blue/green layers. In the embodiment of FIG. 5 the dual axis expansion waveguide display **130** comprises a light source **132** a microdisplay panel **131** and an input image node (IIN) **133** optically coupled to red, green and blue waveguides **134R,134G,134B** each comprise two grating layers. In order that wave guiding can take place in each waveguide the three waveguides are separated by air gaps. In some embodiments the waveguides are separated by a low index material such as a nanoporous film. The red grating layer labelled by R contains an input grating **135R,136R**, a fold grating exit pupil expander **137R,138R** and an output grating **139R,140R**. The grating elements of the blue and green waveguides are labeled using the same numerals with B,G designating blue and red. Since the light paths through the IIN and waveguides in each of the red green and blue waveguides are similar to those illustrated in FIG. 1 they are not shown in FIG. 4. In some embodiments the input, fold and output gratings are all passive, that is non-switching. In some embodiments at least one of the gratings is switching. In some embodiments the input gratings in each layer are switchable to avoid color crosstalk between the waveguide layers. In some embodiments color crosstalk is avoided by disposing dichroic filters **141,142** between the input grating regions of the red and blue and the blue and green waveguides.

(66) In some embodiments a color waveguide based may use just one grating layer in each monochromatic waveguide. In the embodiment shown in FIG. 5, which is similar to the one of FIG. 4, the dual axis expansion waveguide display **150** comprises a light source **132**, a microdisplay panel **131** and an input image node (IIN) **133** optically coupled to red, green and blue waveguides **151R,151G,151B** each comprising one grating layer. The red grating layer labelled by R contains an input grating **152R**, a fold grating exit pupil expander **153R** and an output grating **154R**. The grating elements of the blue and green waveguides are labeled using the same numerals with B,G designating blue and red. Dichroic filters **155,156** are disposed between the input grating regions of the red and blue and the blue and green waveguides to control color crosstalk. Since the light paths through the IIN and waveguides in each of the red green and blue waveguides are similar to those illustrated in FIG. 1 they are not shown in FIG. 5.

(67) FIG. 6 is a plan view of a grating layer **160** in a waveguide display showing a layout comprising an input grating **163**, a fold grating **162** and an output grating **163**. Gratings are shaded using lines showing the orientations the grating fringes with the input grating fringes being aligned at 90 degrees to the X coordinate of the Cartesian reference frame shown, the fold grating fringes at 45 degrees and the output grating fringes at 0 degrees.

(68) In one embodiment of the invention shown in FIG. 7 there is provided an eye tracked display comprising a waveguide display according to the principles of the invention and an eye tracker. In one preferred embodiment the eye tracker is a waveguide device based on the embodiments and teachings of PCT/GB2014/000197 entitled HOLOGRAPHIC WAVEGUIDE EYE TRACKER, PCT/GB2015/000274 entitled HOLOGRAPHIC WAVEGUIDE OPTICAL TRACKER, and PCT Application No.: GB2013/000210 entitled APPARATUS FOR EYE TRACKING. Turning again to FIG. 7 the eye tracked display **170** comprises a dual axis expansion waveguide display based on any of the above embodiments comprising the waveguide **171** containing at least one grating layer incorporating an input fold and output grating, the IIN **173**, the eye tracker comprising the waveguide **173**, infrared detector **174** and infrared source **175**. The eye tracker and display waveguides are separate by an air gap or by a low refractive material. As is explained in the above references, the eye tracker may comprise separate illumination and detector waveguides. The optical path from the infrared source to the eye is indicated by the rays **1033-1035** and the backscattered signal from the eye is indicated by the rays **1036-1037**. The display comprises a waveguide **966** and an input image node **968**. The optical path from the input image node through the display waveguide to the eye box is indicated by the rays **1030-1032**.

(69) In some embodiments of the invention a dual expansion waveguide display further comprises a dynamic focusing element. In some embodiments such as the one shown in FIG. 8 a dual

expansion waveguide display **180** further comprises a dynamic focusing element **181** disposed in proximity to a principal surface of the waveguide display and an eye tracker. Advantageously the dynamic focusing element is a LC device. In some embodiments the LC device combines an LC layer and a diffractive optical element. In some embodiments the diffractive optical element is an electrically controllable LC-based device. In some embodiments the dynamic focusing element is disposed between the waveguide display and the eye tracker. In some embodiments the dynamic focusing element may be disposed in proximity to the surface of the display waveguide furthest from the eye. In some embodiments such as the one illustrated in FIG. **9** the dual expansion waveguide display **190** includes a dynamic focusing element **191** disposed within the IIN. The effect of the dynamic focus device is to provide a multiplicity of image surfaces **1040**. In light field display applications at least four image surfaces are required. The dynamic focusing element may be based on the embodiments and teachings of U.S. Provisional Patent Application No.: 62/176,572 entitled ELECTRICALLY FOCUS TUNABLE LENS. In some embodiment a dual expansion waveguide display further comprising a dynamic focusing element and an eye tracker may provide a light field display based on the embodiments and teachings disclosed in U.S. Provisional Patent Application No.: 62/125,089 entitled HOLOGRAPHIC WAVEGUIDE LIGHT FIELD DISPLAYS.

(70) In some embodiments the waveguide display is coupled to the IIN by an opto-mechanical interface that allows the waveguide to be easily retracted from the IIN assembly. The basic principle is illustrated in FIG. **10A** which shows a dual axis expansion waveguide display **200** comprising the waveguide **201** containing the input grating **202**, fold grating **203** and output grating **204** and the IIN **205**. The apparatus further comprises an optical link **206** connected to the waveguide, a first optical interface **207** terminating the optical link and a second optical interface **208** forming the exit optical port of the IIN. The first and second optical interfaces can be decoupled as indicated by the gap **209** shown in FIG. **10B**. In some embodiments the optical link is a waveguide. In some embodiments the optical link is curved. In some embodiments the optical link is a GRIN image relay device. In some embodiments the optical connection is established using a mechanical mechanism. In some embodiments the optical connection is established using a magnetic mechanism. The advantage of decoupling the waveguide from the IIN in helmet mounted display applications is that the near eye portion of the display be removed when not in used. In some embodiments where the waveguide comprises passive gratings the near eye optics can be disposable.

(71) FIG. **11** illustrates rolled K-vector gratings for use with the invention. Referring first to FIG. **11A**, in some embodiments a rolled K-vector grating **220** is implemented as a waveguide portion containing the discrete grating elements **212-215** having K-vectors **1050-1053**. Referring next to FIG. **11B**, in some embodiments a rolled K-vector grating **220** is implemented as a waveguide portion containing grating elements **222** within which the K-vectors undergoes a smooth monotonic variation in direction including the illustrated directions **1050-1053**.

(72) FIGS. **12A-12D** illustrated front, plan, side and three dimensions views of one eyepiece of a dual axis expansion display used in a motorcycle helmet mount display in one embodiment. The display comprises the waveguide **231**, input grating **232**, fold grating **233**, output grating **234**, a hinge mechanism **235** for attaching the display to the helmet and the waveguide coupling mechanism **236**. FIGS. **13A-FIG. 13B** show three dimensional views of the eyepiece integrated in a motorcycle helmet.

(73) In practical embodiments of the invention care must be taken to ensure that the IIN is optically matched to the waveguide. Waveguides raise optical interfacing issues that are not encountered in conventional optical systems in particular matching the input image angular content to the angular capacity of the waveguide and input grating. The optical design challenge is to match the IIN aperture variation as a function of field angle to the rolled K-vector input grating diffraction direction. Desirably the waveguide should be designed to make the waveguide thickness as small as possible while maximizing the spread of field angles at any given point on the input grating,

subject to the limits imposed by the angular bandwidth of the input grating, and the angular carrying capacity of the waveguide. From consideration of the above description and the teachings of earlier filings such it should be appreciated that coupling collimated angular image content over the full field of view and without significant non-uniformity of the illumination distribution across the pupil requires a numerical aperture (NA) variation ranging from high NA on one side of the microdisplay falling smoothly to a low NA at the other side. For the purposes of explaining the invention the NA is defined as being proportional to the sine of the maximum angle of the image ray cone from a point on the microdisplay surface with respect to an axis normal to the microdisplay. Other equivalent measures may be used for the purposes of determining the most optimal IIN to waveguide coupling. Controlling the NA in this way will ensure high optical efficiency and reduced banding and other illumination non-homogeneities in the case of LED-illuminated displays. Laser-illuminated displays will also benefit from the control of NA variation across the microdisplay particular with regard to homogeneity.

(74) The following embodiments address the problem of varying the NA. In one embodiment using angle selective coatings, gratings wedges, micro elements, freeform elements, etc, inside the IIN. We shall refer to the XYZ Cartesian coordinate system shown in the drawings. For the purposes of explain the invention the Z axis is normal to the waveguide and eyebox planes. The Y and X axes are vertical and horizontal respectively. The rolled-K vectors gratings have their K-vectors tilted in the X-Z plane. Image light in the waveguide propagates substantially in the X-direction. In the following paragraphs we shall describe schemes for varying the NA from high to low (or vice versa) across the horizontal axis of the microdisplay that is along the X-direction. In some embodiments the microdisplay is an LCoS device.

(75) In one embodiment shown in FIG. **14** the IIN **250** comprises a microdisplay panel **251** a spatially-varying NA component **252** and microdisplay optics **253**. The microdisplay optics accepts light **1060** from an illumination source which is not illustrated and deflects the light on to the microdisplay in the direction indicated by the ray **1061**. The light reflected from the microdisplay is indicated by the divergent ray pairs **1062-1064** with NA angles varying along the X axis. In embodiments based on FIG. **14** the spatially-varying NA component is disposed between the microdisplay optics and the microdisplay. In some embodiments the spatially-varying NA component is disposed adjacent the output surface of the microdisplay optics as illustrated in FIG. **15** which shows a microdisplay **261**, illumination component **262**, spatially-varying NA component **263** with input illumination and illumination emitted from the microdisplay optics onto the microdisplay as indicated by the rays **1070,1071**. The light reflected from the microdisplay is indicated by the divergent ray pairs **1072-1074** with NA angles varying along the X axis.

(76) In the embodiments of FIGS. **14-15** the microdisplay is a reflective device. In some embodiments the microdisplay is a transmission device, typically a transmission LCoS device. In the embodiment of FIG. **16** the IIN **260** comprises the backlight **261**, a microdisplay **262** and a variable NA component **263**. Light from the backlight indicated by the rays **1071-1073** which typically has a uniform NA across the backlight illuminates the back surface of the microdisplay and after propagation through the variable NA component is converted into output image modulated light indicated by the divergent ray pairs **1072-1074** with NA angles varying along the X axis.

(77) In some embodiments the principles of the invention may be applied to an emissive display. Examples of emissive displays for use with the invention include ones based on LED arrays and light emitting polymers arrays. In the embodiment of FIG. **17** the IIN **265** comprises an emissive microdisplay **266** and a spatially-varying NA component **267**. Light from the microdisplay indicated by the rays **1077-1079** which typically has a uniform NA across the emitting surface of the display illuminates the spatially-varying NA component and is converted into output image modulated light indicated by the divergent ray pairs **1080-1082** with NA angles varying along the X axis.

(78) The invention does not assume any particular design for the microdisplay optics. In some embodiments the microdisplay optics comprises a polarizing beam splitter cube. In some embodiments the microdisplay optics comprises an inclined plate to which a beam splitter coating has been applied. In some embodiments the microdisplay optics comprises a waveguide device comprising a SBG which acts as a polarization selective beam splitter based on some of the embodiments of U.S. patent application Ser. No.: 13/869,866 entitled HOLOGRAPHIC WIDE ANGLE DISPLAY, and U.S. patent application Ser. No.: 13/844,456 entitled TRANSPARENT WAVEGUIDE DISPLAY. In some embodiments the microdisplay optics contains at least one of a refractive component and curved reflecting surfaces or a diffractive optical element for controlling the numerical aperture of the illumination light. In some embodiments the microdisplay optics contains spectral filters for controlling the wavelength characteristics of the illumination light. In some embodiments the microdisplay optics contains apertures, masks, filter, and coatings for controlling stray light. In some embodiments the microdisplay optics incorporate birdbath optics.

(79) We next describe exemplary embodiments of the spatially-varying NA component. In some embodiments the spatially-varying NA component has a uniformly varying NA characteristic. In some embodiments are provided in a stepwise fashion. It should be various exemplary embodiments of the spatially-varying NA component are illustrative only and, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied.

(80) In some embodiments such as the one illustrated in FIG. 18A a spatially-varying NA component 270 comprises a wedge 271. In some embodiments such as the one illustrated in FIG. 18B a spatially-varying NA component 272 comprises a wedge having a curved surface 273. In some embodiments such as the one illustrated in FIG. 18C a spatially-varying NA component 274 comprises an array of prismatic elements having differing prism angles such as the elements 275, 276. In some embodiments such as the one illustrated in FIG. 18D a spatially-varying NA component 277 comprises an array of lenses having differing apertures and optical powers such as the lens elements indicated by 278, 279.

(81) In some embodiments such as the one illustrated in FIG. 19A a spatially-varying NA component 280 comprises an array of scatter elements such as the elements 281A, 281B providing differing scattered ray angular distributions such as the ones indicated by 282A, 282B respectively. In some embodiments the scattering properties may be provided by surface textures applied to a substrate. In some embodiment the scattering properties may be provided by the bulk properties of the substrate. In some embodiments the substrate may contain particles suspended in a matrix of refractive index differing from that of the particle material. In some embodiments the substrate may be a PDLC material. In some embodiments such as the one illustrated in FIG. 19B a spatially-varying NA component 283 comprises a substrate 284 having a continuously varying scattering characteristics as indicated by the scattered ray angle distributions 285A, 285B. The scattering properties may be provided by surface or bulk medium characteristics.

(82) In some embodiments such as the one illustrated in FIG. 19C a spatially-varying NA component 286 comprises a birefringent substrate 287 having a spatially varying birefringence as represented by the uniaxial crystal index functions 288A, 288B. In some embodiments the substrate provides a continuous variation of birefringence. In some embodiments the substrates comprises discrete elements each have a unique birefringence. In some embodiments a spatially-varying NA component is a scattering substrate with birefringent properties. In some embodiments a spatially-varying NA component is based on any of the embodiments of FIGS. 18A-18D implements using a birefringent substrate. In some embodiments the NA variation across the field is performed using a birefringent layer having comprising a thin substrate coated with a Reactive Mesogen material. Reactive Mesogens are polymerizable liquid crystals comprising liquid crystalline monomers

containing, for example, reactive acrylate end groups, which polymerize with one another in the presence of photo-initiators and directional UV light to form a rigid network. The mutual polymerization of the ends of the liquid crystal molecules freezes their orientation into a three dimensional pattern. Exemplary Reactive Mesogen materials are manufactured by Merck KGaA (Germany).

(83) In some embodiments such as the one illustrated in FIG. 19D a spatially-varying NA component **286** comprises an array of diffractive elements each characterized by a unique K-vector and diffraction efficiency angular bandwidth. For example element **289A** at one end of the component has k-vector $\underline{K}_{\text{sub.1}}$ and bandwidth $\Delta\theta_{\text{sub.1}}$ configured to provide a high NA while element **289B** at the other end has k-vector $\underline{K}_{\text{sub.2}}$ and bandwidth $\Delta\theta_{\text{sub.2}}$ configured to provide a low NA. In some embodiments the grating characteristics vary continuously across the substrate. In some embodiments the gratings are Bragg holograms recorded in HPDLC materials. In some embodiments the gratings are surface relief gratings. In some embodiments the gratings are computer generated diffractive structures such as computer generated holograms (CGHs).

(84) In some embodiments the IIN design addresses the NA variation problem, at least in part, by tilting the stop plane such that its normal vector is aligned parallel to the highest horizontal field angle, (rather than parallel to the optical axis). As illustrated in FIG. 20 the IIN **287** is configured to provide an output field of view of half angle θ defined by the limiting rays **1086**, **1086B** disposed symmetrical about the optical axis **1087**. The stop plane **1088** is normal to the limiting ray **1086B**. It is assumed that the waveguide input grating, which is not illustrated, couples the horizontal field of view into the waveguide (not shown).

(85) As discussed above, in some embodiments such as the one illustrated in FIG. 10 the waveguide display is coupled to the IIN by an opto-mechanical interface that allows the waveguide to be easily retracted from the IIN assembly. FIG. 21A shows a removable near eye display **290** comprising a near eye waveguide component and an IIN. The waveguide component includes an opto-mechanical interface for coupling to the IIN. The waveguide is shown retracted from the IIN assembly. FIG. 21B shows a second 3D view of the HMD **296** with the waveguide component retracted. FIG. 21C shows a 3D view of HMD **297** with the waveguide component and IIN connected and ready for use.

(86) FIG. 22 show schematic front views of two waveguide grating layouts that may be provided by the invention. In the embodiment of FIG. 22A the waveguide **300** comprises a shaped waveguide comprising in a single layer indicated by **1086** an input grating **302**, a fold grating **303** and an output grating **304**. The K-vectors of the three gratings that is the normal vector to the fringes shown inside each grating are indicated by the **1083-1084**. Note that in each case the K-vector is projected in the plane of the drawings. The overall dimensions of the waveguide are 60 mm. horizontal by 47 mm. vertical. In the embodiment of FIG. 22B the waveguide **310** comprises a shaped waveguide comprising in a single layer indicated by **1090** an input grating **313**, a fold grating **314** and an output grating **315**. The K-vectors of the three gratings that is the normal vector to the fringes shown inside each grating are indicated by the **1087-1089**. In each case the K-vector is projected in the plane of the drawings. The overall dimensions of the waveguide are 75 mm. horizontal by 60 mm. vertical. FIG. 23 shows a further general waveguide grating layout that may be provided by the invention. The waveguide **320** comprises a rectangular waveguide comprising in a single layer an input grating **322**, a fold grating **323** and an output grating **324**. The K-vectors of the three gratings that is the normal vector to the fringes shown inside each grating are indicated by the **1091-1093**. In each case the K-vector is projected in the plane of the drawings. The fold grating in this case has Bragg fringes aligned at 45 degrees in the plane of the grating.

(87) FIG. 24 is a 3D illustration of a near display comprising an IIN and waveguide component in one embodiment. [the display **330** comprises an IIN **331**, waveguide **332** containing in a single layer an input grating **33** a fold grating **334** and an output **335**. The waveguide path from entrance pupil **2000** through the input grating, fold grating and output grating and up to the eye box **2005** is

represented by the rays **2001-2004**.

(88) FIG. **25** is a 3D illustration of one embodiment in which there is provided a motorcycle HMD **340** using a near eye waveguide **343** with an opto-mechanical interface **344** for coupling to the IIN **345** which forms part of the helmet. The apparatus is similar to that illustrated in FIG. **21**. FIG. **25A** shows a first operational state **341** in which the waveguide component is fully retracted from the IIN assembly. FIG. **25B** shows the display in its operational state **342** with the waveguide component connected to the IIN.

(89) In some embodiments a dual expansion waveguide display according to the principles of the invention may be integrated within a window, for example, a windscreen-integrated HUD for road vehicle applications. In some embodiments a window-integrated display may be based on the embodiments and teachings disclosed in U.S. Provisional Patent Application No.: 62/125,064 entitled OPTICAL WAVEGUIDE DISPLAYS FOR INTEGRATION IN WINDOWS and U.S. Provisional Patent Application No.: 62/125,066 entitled OPTICAL WAVEGUIDE DISPLAYS FOR INTEGRATION IN WINDOWS. In some embodiments a dual expansion waveguide display may include gradient index (GRIN) wave-guiding components for relaying image content between the IIN and the waveguide. Exemplary embodiments are disclosed in U.S. Provisional Patent Application No.: 62/123,282 entitled NEAR EYE DISPLAY USING GRADIENT INDEX OPTICS and U.S. Provisional Patent Application No.: 62/124,550 entitled WAVEGUIDE DISPLAY USING GRADIENT INDEX OPTICS. In some embodiments a dual expansion waveguide display may incorporate a light pipe for providing beam expansion in one direction based on the embodiments disclosed in U.S. Provisional Patent Application No.: 62/177,494 entitled WAVEGUIDE DEVICE INCORPORATING A LIGHT PIPE. In some embodiments the input image source in the IIN may be a laser scanner as disclosed in U.S. Pat. No. 9,075,184 entitled COMPACT EDGE ILLUMINATED DIFFRACTIVE DISPLAY. The embodiments of the invention may be used in wide range of displays including HMDs for AR and VR, helmet mounted displays, projection displays, heads up displays (HUDs), Heads Down Displays, (HDDs), autostereoscopic displays and other 3D displays.

(90) Some of the embodiments and teachings of this disclosure may be applied in waveguide sensors such as, for example, eye trackers, fingerprint scanners and LIDAR systems.

(91) It should be emphasized that the drawings are exemplary and that the dimensions have been exaggerated. For example thicknesses of the SBG layers have been greatly exaggerated. Optical devices based on any of the above-described embodiments may be implemented using plastic substrates using the materials and processes disclosed in PCT Application No.:

PCT/GB2012/000680, entitled IMPROVEMENTS TO HOLOGRAPHIC POLYMER DISPERSED LIQUID CRYSTAL MATERIALS AND DEVICES. In some embodiments the dual expansion waveguide display may be curved.

(92) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (for example, variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

Claims

1. An optical display, comprising: a first waveguide having a first surface and a second surface; an input coupler; a fold grating; a first output grating and a second output grating overlapping the first output grating; and an input image node (IIN), wherein said input coupler is configured to receive collimated first wavelength image modulated light from said IIN and to cause said first wavelength image modulated light to travel within the first waveguide via total internal reflection between said first surface and said second surface to the fold grating; wherein said fold grating is configured to provide pupil expansion in a first direction and to direct said first wavelength image modulated light to the first output grating and the second output grating via total internal reflection between the first surface and the second surface; and wherein the first output grating is configured to diffract the light from the fold grating into a first angular range and the second output grating is configured to diffract light from the fold grating into a second angular range different from the first angular range.
2. The optical display of claim 1, wherein said IIN comprises a light source, a microdisplay for displaying image pixels and collimation optics, and wherein said IIN projects the image displayed on said microdisplay such that each image pixel is converted into a unique angular direction within said first waveguide.
3. The optical display of claim 1, further comprising a second waveguide comprising a first surface and a second surface, an input coupler, a fold grating, and an output grating, wherein the input coupler of the second waveguide is configured to receive collimated second wavelength image modulated light from said IIN.
4. The optical display of claim 3, further comprising a dichroic filter disposed between the input coupler regions of said first and second waveguides.
5. The optical display of claim 1, wherein each of said input coupler, said fold grating, said first output grating, and said second output grating is a grating selected from the group of a grating recorded in a liquid crystal and polymer material system, a Bragg grating, a switchable Bragg grating, a grating recorded in a liquid crystal and polymer material system and a surface relief grating.
6. The optical display of claim 1, wherein said input coupler is a grating or a prism.
7. The optical display of claim 1, wherein said first waveguide further comprises a first optical interface; and said IIN further comprises a second optical interface, wherein said first and second optical interface can be decoupled by one of a mechanical mechanism or a magnetic mechanism.
8. The optical display of claim 1, wherein at least one of said input coupler, said fold grating, said first output grating, or said second output grating is a rolled k-vector grating.
9. The optical display of claim 1, wherein said first surface and said second surface are curved.
10. The optical display of claim 1, wherein said IIN comprises a laser scanner for projecting image modulated first wavelength light over an angular range.
11. The optical display of claim 1, wherein said optical display provides one of a HMD, a HUD, an eye-slaved display, a dynamic focus display or a light field display.
12. The optical display of claim 1, wherein at least one of said input coupler, fold grating, first output grating, or second output grating multiplexes at least one of color or angle.
13. The optical display of claim 1, further comprising at least one selected from the group of: an eye tracker, a dynamic focus lens disposed within said IIN, a dynamic focus lens disposed in proximity to the first set of surfaces of the first waveguide, a beam homogenizer, a quarter wave plate applied to a surface of said waveguide, and at least one gradient index image transfer waveguide portion.
14. The optical display of claim 1 wherein said IIN further comprises a spatially varying numerical aperture component for providing a numerical aperture variation along at least one direction.

15. The optical display of claim 14 wherein said spatially varying numerical aperture component has at least one characteristic selected from the group of diffractive, birefringent, refracting and scattering.
 16. The optical display of claim 1, wherein said IIN further comprises at least one diffractive optical element.
 17. The optical display of claim 1, wherein said waveguide includes at least one plastic substrate.
 18. The optical display of claim 1, wherein at least one said input coupler, fold grating, first output grating, or second output grating has spatially varying characteristics.
 19. The optical display of claim 1 integrated within a window.
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